

Revisiting Fundamentals of the European Gas Market: The Role of Supply Substitution

Jorge Arenas Amoros*

This version: September 23, 2026

Abstract

The 2022 European energy crisis exposed the central role of supply substitution in natural gas markets, as disruptions to Russian pipeline flows were met by a sharp surge in liquefied natural gas (LNG) imports. This paper revisits the fundamentals of the European gas market by explicitly distinguishing between pipeline gas and LNG supplies. Using a Bayesian structural vector autoregression identified through sign and elasticity restrictions, the analysis jointly identifies the short-run price elasticities of each supply channel, of consumption and of inventories. The results reveal little heterogeneity between the two supply channels at monthly frequency: both respond weakly to the price, and the flexibility of LNG operates through the reallocation of cargoes over several months rather than through immediate arrivals. What moves the price of gas is the demand for storage, which accounts for close to half of its variance, in calm years as well as in crises. The estimates attribute the 2021–2022 crisis mainly to storage accumulation, layered on a smaller but far more persistent loss of pipeline supply, and show the 2026 Hormuz episode moving with news rather than volumes, the price rising when the strait closed and falling when the missing cargoes failed to arrive. Counterfactual scenarios of the 2022 crisis quantify the stabilising role of demand adjustment, which contained the price about three times as much as LNG did. Supply composition, storage and expectations, rather than total import volumes alone, shape European gas prices.

Keywords: natural gas; liquefied natural gas; structural VAR; price elasticities; storage; European energy crisis.

JEL classification: C11, C32, Q41, Q43, Q48.

*Universidad de Alicante. Email: jorgearenasamoros@gmail.com. I am grateful to Christiane Baumeister, Hilde C. Bjørnland, Lutz Kilian and Stefano Fasani for helpful comments and discussions, and to participants at the 2026 conference of the International Association for Applied Econometrics (IAAE) and at the 2026 International Conference on Macroeconomic Analysis and International Finance (ICMAIF). FUNDING ACKNOWLEDGEMENT PENDING. All remaining errors are my own.

1 Introduction

The energy crisis experienced by the European Union in 2022 reignited the debate on the role of fossil fuels in modern economies and, in particular, on the vulnerabilities associated with energy dependence. Natural gas accounts for roughly a quarter of total energy consumption in the European Union and is predominantly extracted outside Europe, so unexpected disruptions to gas production or transportation in distant regions can have large and immediate effects on European energy prices and macroeconomic conditions. While oil markets have long been the focus of extensive empirical research, the modelling of natural gas markets has gained prominence only recently, reflecting the growing importance of gas as an input for industrial production, electricity generation and household heating, and the rising attention of policymakers and central banks in the context of inflationary pressures and energy security.

Natural gas markets also differ from crude oil markets in structure. Whereas crude oil is globally integrated and predominantly shipped by sea, natural gas markets have historically been fragmented across regions, a fragmentation that largely reflects transportation constraints (Szafranek and Rubaszek, 2024). Gas reaches Europe through two physically and contractually distinct delivery channels, pipeline gas and liquefied natural gas (LNG). Pipeline gas has traditionally been the dominant source, accounting for approximately 85% of total net imports until 2018; since 2019 its share has declined markedly, falling below 60% by 2024. The increasing role of LNG has fundamentally altered the structure of European gas supply, introducing a more flexible but capacity-constrained source that ties the European market to global conditions.

Despite these structural differences, much of the empirical literature on natural gas markets adopts structural vector autoregressive (SVAR) frameworks developed for crude oil by Kilian (2009); Kilian and Murphy (2014); Baumeister and Hamilton (2019), in which supply enters as a single aggregate. For crude oil that is a reasonable approximation, because extraction is the only channel through which supply can respond to the price. For natural gas the abstraction is problematic: it ignores the heterogeneity across supply channels and the scope for substitution between them, and it discards the information embedded in shifts in the composition of imports (Moll et al., 2023; Hamilton, 2023). The 2022 crisis illustrates the point. Within a few months of the Russian invasion of Ukraine, deliveries through the Russian pipeline network collapsed from more than 40% of total European net imports to a residual share, while LNG imports rose by over 60% to partly absorb the shortfall. When supply is treated as a single aggregate, this reallocation appears as a modest change in total imports, masking the adjustment occurring beneath the surface. The episode also raises a question that an aggregate model cannot answer: how much of the rise in the price of gas was attributable to the loss of pipeline supply itself, and how much to the limited capacity of LNG to expand in its place.

This paper models the European natural gas market by explicitly distinguishing between pipeline gas and LNG supplies, and it contributes to the literature in two ways. The first contribution is to identify the short-run price elasticities of the two supply channels separately, which to our knowledge no previous study does for Europe: existing applications either treat supply as a single aggregate or, where they distinguish sources, do not identify

a separate price elasticity for each. Estimating the two elasticities jointly avoids the implicit assumption that pipeline gas and LNG interact with prices, activity and storage in an identical manner, an assumption at odds with a substantial institutional literature. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay and destination clauses and is constrained by fixed infrastructure with limited spare capacity, while LNG is increasingly traded on shorter-duration spot or hub-indexed contracts, with cargoes redirectable across regions in response to price differentials (Sharples, 2023; Ritz, 2019; Molnar, 2022). The separation is what makes it possible to test whether LNG behaves as flexibly as that literature describes, and to trace the two channels’ distinct transmission to prices, inventories and consumption.

The second contribution is one of identification, and it arises directly from the first. Following the oil literature, the demand equation in this class of models is written for a quantity aggregate constructed as supply minus the change in inventories. With two supply channels that construction leaves the demand and LNG-supply price elasticities identified only through their difference, and because the framework is explicitly set-identified, prior information does not wash out asymptotically (Baumeister and Hamilton, 2015), so a longer sample cannot resolve the split. To our knowledge this consequence has not previously been made explicit, and it matters for any disaggregated application of this framework, not only for ours. It is the reason observed gas consumption enters our model: with the demand equation written for a quantity that is actually measured, the demand elasticity is identified in its own right, and once demand is pinned down, so is the elasticity of LNG supply.

The framework is the Bayesian structural VAR of Baumeister and Hamilton (2015, 2019), identified through sign and elasticity restrictions, in which informative priors over the contemporaneous structural matrix combine with the likelihood to deliver set-identified parameters. To our knowledge this identification strategy has not previously been applied to the European gas market; related frameworks have been applied to the United States market by Rubaszek et al. (2021) and Farag (2024). Its advantage in this setting is that priors on the contemporaneous elasticities remain informative even asymptotically, which matters in energy markets, where short-run elasticities are difficult to estimate from the data alone. The model is estimated on monthly data covering the vast majority of the European Union’s gas market, from February 2014 to May 2026.

The estimates change the prevailing view of the European gas market in three ways. First, the two supply elasticities turn out to be small and similar in size, with LNG if anything slightly the more responsive of the two, though the ordering is not established. The institutional case for treating the two channels as distinct remains, and our impulse responses show them transmitting differently, but the data do not support quantifying that distinction as a large gap between contemporaneous price elasticities. Second, the price-elastic side of the market is storage. Inventories respond to the price with a well-identified negative elasticity, while the price elasticity of consumption, estimated here for the first time against an observed rather than a constructed quantity, is close to zero, as monthly demand driven by weather and heating requirements should be. Precautionary demand for inventories, as a consequence, explains more of the variance of the price of gas than any other shock at every horizon. Third, the impulse responses add the time dimension. A precautionary storage shock moves

the price more than any other disturbance on impact and is the one that unwinds the most, shedding a third of its peak effect within two years, while shocks to either physical supply channel build more slowly and shift the price level almost permanently. Which channel is at work in a given episode therefore determines how long its price effects should be expected to last.

The 2026 closure of the Strait of Hormuz falls inside the estimation sample, so the two European gas crises of the last five years can be read with a single identified model. The 2021–2022 crisis emerges as a crisis of storage demand superimposed on a persistent pipeline loss: precautionary accumulation drives the August 2022 peak and then dissipates, the pipeline shortfall sustains the price level that remains, and LNG acts as a partial buffer rather than a source of the shock. The 2026 episode reads as an expectational one, the price moving with the news rather than with the cargoes. The counterfactuals point the same way: removing the autonomous reduction in gas demand over the window of Council Regulation 2022/1369 raises the counterfactual price by roughly three times as much as either LNG scenario moves it, and with the LNG channel closed off entirely, consumption would have been essentially unchanged, the shortfall coming out of storage. The scenario in which LNG replaces every missing pipeline unit would, in addition, have required import volumes beyond the regasification and shipping capacity available in 2022–2023, which marks the limits of supply flexibility. The European response to the 2022 crisis, on these estimates, rested more on the reduction of demand and on the buffering role of storage than on the expansion of LNG receipts.

The macroeconomic stakes are first-order. Natural gas was the dominant proximate driver of the surge in euro-area HICP from 5.1% in January 2022 to a peak of 10.6% in October 2022, and the European Central Bank’s response was the fastest tightening cycle in its history. The pass-through from gas prices to euro-area consumer prices behind these developments is by now well documented (Adolfson et al., 2024; Alessandri and Gazzani, 2025; Boeck and Zörner, 2025; Casoli et al., 2024; López et al., 2024). The horizon structure just described bears directly on that calibration: a gas-price spike driven by storage demand is the more likely to unwind, as the 2026 episode illustrates, while one driven by a physical loss of supply shifts the price level durably and is the more likely to feed into medium-run inflation and into expectations.

The remainder of the paper is organised as follows. Section 2 reviews the related literature on energy-market SVARs and on the contractual and infrastructural differences between pipeline gas and LNG. Section 3 presents the econometric methodology of the Baumeister–Hamilton framework. Section 4 describes the data, sets out the identification problem created by constructed quantity aggregates and the six-variable specification that addresses it, and states the priors. Section 5 reports the posterior estimates, impulse responses, variance and historical decompositions, the counterfactual exercises for 2022–2023, and the joint anatomy of the two crises. Section 6 concludes.

2 Literature Review

This section relates the paper to four strands of literature. The first is the structural VAR analysis of energy markets, which supplies the framework and, as we argue below, the identification problem that motivates our specification. The second is the theory of storage and the treatment of inventories in commodity models, which is where our central empirical result belongs. The third is the SVAR literature on natural gas specifically. The fourth is the work on the contractual and infrastructural differences between pipeline gas and LNG, which motivates separating the two supply channels.

Structural VARs of energy markets and the identification problem

The modern empirical analysis of energy price shocks originates in the crude oil literature. Hamilton (1996) documented a strong negative relationship between oil price increases and U.S. macroeconomic activity, and Kilian (2009) proposed the structural VAR that has organised the field since, distinguishing oil supply shocks, global aggregate demand shocks and oil-specific demand shocks driven by precautionary or speculative motives. The central lesson, that a price increase has different macroeconomic consequences depending on its source, carries directly to natural gas.

General treatments of the identification problem in this class of models are given by Lütkepohl and Kilian (2017). Kilian and Murphy (2012, 2014) showed that purely agnostic sign restrictions are too weak to discipline this class of models and argued for bounds on short-run elasticities and for the explicit use of inventory data. Baumeister and Hamilton (2015, 2019) formalised the argument in a Bayesian framework: informative priors over the contemporaneous structural matrix combine with the likelihood to deliver set-identified parameters, and the priors, being part of the identification, do not wash out as the sample grows. That property is what makes the present paper’s methodological point consequential rather than pedantic. A parameter about which the likelihood is uninformative will be reported back to the researcher at its prior location no matter how long the sample, and the posterior credible interval will not reveal that this has happened. Diagnosing such cases requires examining what combination of parameters the likelihood actually speaks to, which is the exercise we perform in Section 4.3.

The specific difficulty we identify concerns the construction of the quantity aggregate on which the demand equation is imposed. In the oil applications the corresponding aggregate is production minus the change in inventories, and the single supply variable that enters it also appears as the dependent variable of the supply equation. In gas applications with two supply channels the same construction places one supply variable on both sides of the system, and the demand and supply elasticities of that channel become identified only through their difference. To our knowledge this consequence has not been stated explicitly, though it is closely related to two known concerns: the sensitivity of set-identified SVARs to prior placement emphasised by Baumeister and Hamilton (2015), and the measurement problems that arise when a model variable is constructed from others rather than observed. Kilian and Lee (2014) make a related observation from the opposite direction, showing how much the inference on speculative demand in oil depends on the quality and coverage of the inventory

series used to construct it.

Storage, precautionary demand and the theory of price of storage

The result that survives most robustly in our estimates concerns inventories, which places the paper in a literature older than the SVAR tradition. Working (1949) established that the price relationship between a commodity available now and the same commodity available later is governed by the convenience yield of holding stock, so that inventory behaviour and the spot price are jointly determined rather than sequentially so. Deaton and Laroque (1992) formalise the asymmetry that matters here, and it is worth stating their result precisely: speculative storage smooths prices while stocks are held, and spikes arise when inventories run out and the buffer is absent, sensitivity to shocks is greatest exactly when stocks are low. They are also candid that the estimated storage model cannot reproduce the high autocorrelation of observed commodity prices, a limitation they emphasise themselves. The forward-looking channel is supplied by Pindyck (2001) and Alquist and Kilian (2010): because the spot price and the marginal convenience yield are jointly determined, an increase in uncertainty about future supply raises the spot price immediately, with inventories predetermined on impact and the price overshooting before stocks have moved at all. That is the mechanism our 2026 evidence displays in the data.

In the energy SVAR literature this mechanism enters as precautionary or speculative demand: as a residual demand shock interpreted through expectations of future supply shortfalls in Kilian (2009), whose model contains no inventory data, and identified from the joint behaviour of inventories and the price in Kilian and Murphy (2014). The oil evidence assigns the channel a modest share of price variance, 9% at long horizons in Kilian and Murphy (2014), and Baumeister and Hamilton (2019) revise it further downward after allowing for measurement error in global oil inventories, most of whose monthly variation their estimates attribute to noise. For the U.S. gas market, estimates in the same Bayesian framework place the storage-demand share of price variance at roughly 12–19% (Rubaszek et al., 2021) and 28–32% (Frag, 2024), second in both cases to flow demand. Our estimates assign it a larger role still in European gas: precautionary storage demand accounts for more than half of the impact variance of the real gas price and around 40% at horizons beyond a year, and the contemporaneous elasticity of inventories to price is large and sharply identified. The finding is consistent with the institutional structure of the European market, in which seasonal storage is large relative to monthly consumption, on our estimates around 1.8 months, and in which filling trajectories became an explicit object of policy after 2022. It also rationalises the two crisis episodes we examine: both were priced primarily through storage, in 2021–2022 because stocks entered the winter unusually low, and in 2026 because a chokepoint closure is precisely the kind of news about future availability to which the theory says a storage-constrained market responds. Two further differences make the contrast with the oil evidence interpretable rather than anomalous. European gas storage is directly and well measured, national closing-stock reports across 22 countries, cross-checkable against daily transparency data, with the balance residual of Section 4.2 confirming internal consistency, so the measurement problem that leads Baumeister and Hamilton (2019) to discount the channel in oil does not arise with comparable force here. And the price used is a spot hub

price, on which expectations are priced within days; we return to this point, and to the quantitative contrast with Güntner et al. (2024), in Section 5.1.

Structural VARs of natural gas markets

European gas markets have historically been regionally fragmented owing to transportation constraints and infrastructure specificity, with supply relying on pipeline imports under rigid long-term contracts. Early empirical work focused on the linkage between gas and oil prices under oil-indexed contracts; Szafranek and Rubaszek (2024) document that this relationship weakened markedly from the mid-2000s onward with liberalisation and the transition to hub-based pricing. Recent SVAR analyses of the European market identify supply shocks, activity- or weather-driven demand shocks and storage-driven precautionary demand, and generally find inelastic short-run demand alongside constrained supply responses. Güntner et al. (2024) show for Germany that the collapse in Russian pipeline deliveries operated as a dominant negative supply shock partly offset by LNG inflows, with precautionary storage demand amplifying the price response; Adolfsen et al. (2024) trace the pass-through of gas-market disturbances to euro-area inflation, and Boeck and Zörner (2025) document that part of that pass-through operates through inflation expectations. Alessandri and Gazzani (2025) identify European gas supply shocks with an instrument built from daily gas-market news and find them markedly stagflationary for the euro area, while Casoli et al. (2024) and López et al. (2024) quantify the pass-through of gas prices to euro-area consumer prices. Closest to our supply-side question, Adolfsen et al. (2026) distinguish pipeline from LNG supply shocks in a weekly model of the European market and include the spread between the Asian and the European gas price among the endogenous variables, so that global LNG conditions enter the system directly. They find that the two supply shocks move prices and inventories similarly, and that precautionary demand played a larger role in the 2021–2023 crisis than previously recognised. Our monthly framework reaches both conclusions independently, and adds what a shock decomposition alone cannot deliver, the price elasticity of each supply channel. Moll et al. (2023) and Hamilton (2023) argue that aggregating pipeline gas and LNG into a single supply measure conceals compositional adjustment and understates the role of supply-side forces.

The U.S. market provides a comparator with different structure. It is largely self-sufficient, with price formation dominated by domestic market-based production, and the SVAR literature there separates domestic supply shocks, in particular those associated with the shale-gas expansion, from weather, activity and storage demand. Rubaszek et al. (2021) estimate short-run demand elasticities larger in absolute value than those typically found for oil, attributing this to fuel-switching in power generation, and document a non-negligible role for storage-related demand. Farag (2024) examine the growing international exposure of U.S. prices following the expansion of LNG export capacity. Our own demand elasticity, estimated against observed European consumption, is considerably smaller than these U.S. estimates, which is what the different composition of European gas use, weighted toward space heating, and the different frequency of measurement would lead one to expect.

Pipeline gas and LNG as distinct supply channels

A substantial literature establishes that the two channels differ in contractual and physical terms. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay and destination clauses and is constrained by fixed infrastructure with limited spare capacity, implying low short-run flexibility (Molnar, 2022; Oxford Institute for Energy Studies, 2022). LNG has become more flexible as contract durations have shortened, destination restrictions have weakened and spot trading has expanded, so that cargoes can be redirected across regions in response to price differentials (Sharples, 2023; Ulvestad et al., 2012). The broader question of which parts of a gas system supply flexibility, and at what horizon, is treated by International Energy Agency (2002), and Sedlar et al. (2023) examines the contribution of LNG receiving capacity to market flexibility in a European setting. Ritz (2019) argues that this flexibility gives LNG strategic weight beyond its volume share by providing a credible outside option that constrains the market power of pipeline exporters. The 2022 experience is consistent with the mechanism: LNG imports rose sharply as Russian pipeline flows collapsed, though only at exceptionally high spot prices.

Our contribution to this strand is partly negative. The institutional case for treating the two channels as economically distinct is strong, and we retain the distinction in the model. What our estimates do not support is the further step of quantifying that distinction as a large gap between contemporaneous price elasticities, at least at monthly frequency: once the LNG-supply elasticity is identified from a moment it does not share with the demand equation, the posterior leans toward the LNG elasticity being the larger of the two without establishing the ordering. Appendix P shows in addition that the estimated gap depends on which point of the LNG chain the data measure, arrivals of cargoes or send-out from terminals, which suggests that quantities of this kind are better read as properties of a measurement concept than as structural constants of the market. The flexibility of LNG is real, but at monthly frequency it appears to operate through the reallocation of volumes over horizons longer than a month rather than through a large contemporaneous response to the European price.

3 Methodology

The empirical model is a structural vector autoregression (SVAR) for six monthly variables of the European gas market: pipeline supply, industrial production, the real price of gas, storage, LNG arrivals and gas consumption. Its purpose is to recover the short-run price elasticities of the two supply channels and of demand, together with the structural shocks that drive the price. This section presents the econometric framework in general notation; Section 4.2 specifies the equations, and with them the economic content of the system. In matrix form the model reads

$$\mathbf{A}y_t = \mathbf{B}x_{t-1} + u_t \quad (1)$$

$$u_t \sim N(0_{n \times 1}, \mathbf{D}) \quad (2)$$

where $y_t = (y_{1t}, y_{2t}, \dots, y_{nt})'$ is the $n \times 1$ vector of endogenous variables, $\mathbf{A}$ is the $n \times n$ matrix that contains all the contemporaneous elasticities across the endogenous variables,

$x_{t-1} = (y_{t-1}, \dots, y_{t-m})'$ is the $k \times 1$ vector of lagged regressor, $\mathbf{B}$ is the $n \times k$ autoregressive matrix of coefficients and $\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$ is the variance of the structural innovations.

But it also has a reduced form, which can be interpreted as a conventional VAR:

$$y_t = \Phi x_{t-1} + \varepsilon_t \quad (3)$$

$$\varepsilon_t \sim N(0_{n \times 1}, \Omega) \quad (4)$$

$$\Phi = A^{-1}B \quad (5)$$

$$\Omega = A^{-1}D(A^{-1})' \quad (6)$$

The model is estimated using Bayesian methods with sign restrictions. In set-identified models, the prior of the contemporaneous elasticities does not vanish asymptotically, as Baumeister and Hamilton (2015) (henceforth BH) demonstrated. The methodology was subsequently extended and applied to the global crude oil market by Baumeister and Hamilton (2019) and to the U.S. natural gas market by Rubaszek et al. (2021) and Farag (2024). The BH framework offers the flexibility to center all priors of the elements in $\mathbf{A}$ at values supported by the literature. Anchoring the location and scale of the prior for the structural elasticities at empirically grounded values lets the model incorporate information that is not available from the data alone. In this section we provide a brief description of the methodology.

3.1 Prior

The starting point for setting the prior is decomposing $p(A, B, D)$ as:

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D}) = p(\mathbf{A}) \times p(\mathbf{D} \mid \mathbf{A}) \times p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}) \quad (7)$$

The prior for the diagonal variance-covariance matrix $\mathbf{D}$ consists in setting that the inverse of its elements d_{ii}^{-1} are jointly independent and follow a Gamma distribution with $\kappa_i, \tau_i(\mathbf{A})$ as parameters.

$$p(\mathbf{D} \mid \mathbf{A}) = \prod_{i=1}^n p(d_{ii} \mid \mathbf{A}) \quad (8)$$

$$d_{ii}^{-1} \mid \mathbf{A} \sim \Gamma(\kappa_i, \tau_i(\mathbf{A})), \quad (9)$$

Similarly, rows of matrix $\mathbf{B}$ are assumed to be jointly independent and to follow a multivariate normal distribution with parameters m_i, M_i .

$$p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}) = \prod_{i=1}^n p(b_i \mid \mathbf{D}, \mathbf{A}) \quad (10)$$

$$b_i \mid \mathbf{A}, \mathbf{D} \sim N(m_i, d_{ii}M_i), \quad (11)$$

The prior for $\mathbf{A}$ can be freely chosen to reflect any knowledge about the structural relations between the endogenous variables. In the case of this study it will be the joint distribution of several truncated Student-t.

3.2 Likelihood

Innovation of the structural model is assumed to be normal.

$$p(Y_T | \mathbf{A}, \mathbf{D}, \mathbf{B}) = (2\pi)^{-Tn/2} |\det(\mathbf{A})|^T \det(\mathbf{D})^{-T/2} \quad (12)$$

$$\times \exp \left\{ -\frac{1}{2} \sum_{t=1}^T (\mathbf{A}y_t - \mathbf{B}x_{t-1})' \mathbf{D}^{-1} (\mathbf{A}y_t - \mathbf{B}x_{t-1}) \right\}, \quad (13)$$

3.3 Posterior

To proceed with the posterior distribution, we need a similar decomposition as before:

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D} | Y_t) = p(\mathbf{A} | Y_t) \times p(\mathbf{D} | \mathbf{A}, Y_t) \times p(\mathbf{B} | \mathbf{A}, \mathbf{D}, Y_t) \quad (14)$$

where the conditional posterior of the elements of the diagonal variance-covariance matrix, $p(\mathbf{D} | \mathbf{A}, Y_t)$, is expressed as the joint distribution of all elements on the diagonal:

$$p(\mathbf{D} | \mathbf{A}Y_t) = \prod_{i=1}^n p(d_{ii} | \mathbf{A}Y_t) \quad (15)$$

$$d_{ii}^{-1} | \mathbf{A} \sim \Gamma(\kappa_i^*, \tau_i^*(\mathbf{A})), \quad (16)$$

The parameters κ_i^* and $\tau_i^*(\mathbf{A})$ that fully characterize this posterior are functions of the data and of $\mathbf{A}$, as follows:

$$\kappa_i^* = \kappa_i + T/2 \quad (17)$$

$$\tau_i^*(\mathbf{A}) = \tau_i(\mathbf{A}) + \xi_i^*(\mathbf{A}) \quad (18)$$

And the term ξ_i^* refers to the sum of squared residuals from regressing $\tilde{Y}_i(\mathbf{A})$ on $\tilde{X}_i$ defined as:

$$\xi_i^*(\mathbf{A}) = (\tilde{Y}_i(\mathbf{A})' \tilde{Y}_i(\mathbf{A})) - (\tilde{Y}_i(\mathbf{A})' \tilde{X}_i) (\tilde{X}_i' \tilde{X}_i)^{-1} (\tilde{X}_i' \tilde{Y}_i(\mathbf{A})) \quad (19)$$

$$\tilde{Y}_i = [y_1' a_i \quad \cdots \quad y_T' a_i \quad m_i' P_i]'_{(T+k) \times 1} \quad (20)$$

$$\tilde{X}_i = [x_0' \quad \cdots \quad x_{T-1}' \quad P_i]'_{(T+k) \times k} \quad (21)$$

where $M_i = P_i' P_i$ and a_i are the rows of the matrix $\mathbf{A}$.

The conditional posterior of the autoregressive matrix, $p(\mathbf{B} | \mathbf{A}, \mathbf{D}, Y_t)$, can be defined as the joint posterior of its rows:

$$p(\mathbf{B} | \mathbf{A}, \mathbf{D}, Y_t) = \prod_{i=1}^n p(b_i | \mathbf{D}, \mathbf{A}) \quad (22)$$

$$b_i | \mathbf{A}, \mathbf{D}Y_t \sim N(m_i^*, d_{ii} M_i^*) \quad (23)$$

where

$$m_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1} (\tilde{X}_i' \tilde{Y}_i), \quad (24)$$

$$M_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1}. \quad (25)$$

Finally, the posterior of $\mathbf{A}$ is:

$$p(\mathbf{A} \mid \mathbf{Y}_T) = \frac{k_T p(\mathbf{A}) [\det(\mathbf{A} \hat{\Omega}_T \mathbf{A}')]^{T/2}}{\prod_{i=1}^n [(2\tau_i^*/T)\kappa_i^*]} \prod_{i=1}^n \left\{ \frac{|\mathbf{M}_i^*|^{1/2}}{|\mathbf{M}_i|^{1/2}} \frac{\tau_i^{\kappa_i}}{\Gamma(\kappa_i)} \Gamma(\kappa_i^*) \right\}.$$

where $\hat{\Omega}_T = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_t \hat{\varepsilon}_t'$ is the sample variance covariance matrix estimated by maximum likelihood.

Inference draws from $p(\mathbf{A} \mid \mathbf{Y}_T)$ using a Random-Walk Metropolis–Hastings sampler. The proposal density is a multivariate Gaussian with covariance $\xi \cdot \hat{\mathcal{H}}^{-1}$, where $\hat{\mathcal{H}}$ is the Hessian of the log-posterior evaluated at the mode and the scaling factor is fixed at $\xi = 0.0625$. We run 2×10^6 draws with a burn-in of 1.7×10^6 , retaining the remaining 3×10^5 post-burn-in draws for inference. Cumulative impulse responses, forecast-error variance decompositions, historical decompositions, and the counterfactual scenarios in Section 5.5 are computed for each retained draw and summarised by their pointwise posterior median and 5–95% quantiles. The random seed is fixed at 140778 for reproducibility. Convergence is assessed by visual inspection of the trace plots of the contemporaneous elasticities and by comparing the prior–posterior marginals reported in Section 5.1.

4 Empirical model and data

4.1 Data

We model the European natural gas market as a six-variable system. Five variables describe the market as it is usually specified in this literature: two supply channels, real activity, the real price of gas and inventories. The sixth variable, observed gas consumption, is what distinguishes the specification used here from its predecessors. Section 4.3 explains why that sixth variable is the pivot of the identification strategy; we first describe what each series measures.

The first supply channel, denoted q_{pip} , comprises net pipeline imports plus EU domestic production. Both share contractual and physical rigidities that prevent rapid adjustment within a month. Pipeline gas is governed by long-term, route-specific contracts with take-or-pay clauses, fixed transport infrastructure and limited spare capacity (Molnar, 2022). Domestic production is constrained by reservoir decline curves and regulatory frameworks that operate over multi-year horizons; the gradual phase-out of the Groningen field illustrates how the latter, rather than monthly price dynamics, drives variation in this component. We aggregate the two because they share the relevant economic property for the question at hand, near-zero short-run price elasticity, which makes them the slow-adjusting component of European gas supply.

The second channel, q_{LNG} , comprises net LNG imports. LNG operates as the flexible component of European supply: cargoes are redirectable across regions in response to price differentials, contract durations have shortened, destination clauses have weakened, and exposure to spot prices has expanded (Sharples, 2023; Ritz, 2019). The 2022 energy crisis illustrates this asymmetry sharply: in response to the collapse in Russian pipeline deliveries,

EU LNG imports surged by over 60% within months, while pipeline-and-domestic supply showed virtually no compensating expansion in the aggregate. Our data record LNG as it arrives aboard ocean tankers, on a regasified-equivalent basis, so the variable measures cargo receipts rather than send-out from the terminals; the buffer that separates the two, and the reasons for keeping pipeline imports and domestic production in a single aggregate, are discussed in Appendix B.

The sixth variable, c , is observed EU gas consumption: Eurostat’s inland consumption of natural gas for the EU27 aggregate, in millions of cubic metres. This is a directly measured quantity, reported monthly by member states and aggregated by Eurostat, and it is the variable that the demand equation of the model is about. It is the total inland consumption of the economy, covering households, industry, services and gas used in power and heat generation, and it is Eurostat’s observed variant, reported from metered deliveries, as distinct from the calculated variant that Eurostat derives residually from supply and stock changes; using the latter would reintroduce the construction problem of Section 4.3. It enters the system in log growth rates and is seasonally adjusted with X-13 ARIMA-SEATS at the level stage, exactly as the pipeline, LNG and inventory series are; we verify in Section 5.1 that the alternative treatment, leaving the series unadjusted and absorbing seasonality through monthly dummies, leaves every reported result unchanged.¹

The remaining three series are as follows. Total natural gas inventories s are taken from JODI; the variable is included because of its central role in consumption smoothing and in precautionary or speculative behaviour. Industrial production ip is the manufacturing index for the euro area, sourced from Eurostat, seasonally and calendar adjusted. The real price of natural gas p is the Dutch TTF benchmark from the World Bank Commodity Price Database (*Natural gas, Europe*, Pink Sheet monthly series), deflated by the euro-area HICP all-items index published by Eurostat (2015=100).² The two supply aggregates are constructed from the JODI-Gas database, summing reported pipeline imports, LNG imports and domestic production across 22 EU member states.³

The vector of endogenous variables is therefore

$$y_t = [q_{pip}, ip, p, s, q_{LNG}, c]' \quad (26)$$

with all series expressed in logarithmic growth rates (percentage points per month) to remove unit roots in prices, industrial production and quantities. The sample spans February 2014

¹The two adjusted growth series correlate at 1.000 with the same standard deviation (6.86 percentage points), differing only in how the January seasonal factor is apportioned (± 0.66 pp); the crisis movements that carry the identification are untouched (March 2022 +8.4, March 2026 −7.9).

²Eurostat migrated the HICP from ECOICOP version 1 to version 2 during 2026; the legacy `prc.hicp.midx` series is frozen at December 2025 and the current all-items index is published under `prc.hicp.minr`. The two are the same index on the same base and concatenate without rescaling. Because the price enters the system in log growth rates, the choice of index base is irrelevant to the estimates by construction.

³Country panel: AT, BE, CZ, DE, DK, EE, ES, FI, FR, GR, HU, IE, IT, LT, LU, LV, NL, PL, PT, SE, SI, SK. Five EU members are excluded due to incomplete JODI coverage: BG, HR, MT and RO have data only from 2019 onwards, and CY does not report. The 22 countries jointly account for approximately 95% of EU-27 natural gas consumption. A robustness check using IEA *Gas Trade Flows* data in place of JODI for the two supply aggregates is reported in Appendix P.

through May 2026, for a total of 148 monthly observations, and the VAR is estimated with 6 monthly lags, consistent with the lag choice in Baumeister and Hamilton (2019) and Rubaszek et al. (2021); impulse responses are reported to a horizon of 24 months.⁴

The sample start is dictated by the consumption series, which Eurostat publishes from January 2014; without it the model could be estimated from February 2010. We accept the shorter sample because the two are not of comparable value. The additional pre-2014 observations would sharpen the precision of coefficients that are already signed and small, whereas the consumption variable delivers the separate identification of two elasticities that the constructed-aggregate specification cannot tell apart at any sample length, as Section 4.3 explains.

4.2 Specification and assumptions

It is useful to see the system as a whole before deriving its equations one by one. In matrix form the model is

$$\begin{pmatrix} 1 & 0 & -\gamma_{qp} & 0 & 0 & 0 \\ 0 & 1 & -\alpha_{yp} & 0 & 0 & 0 \\ 0 & -\alpha_{qy} & -\alpha_{qp} & 0 & 0 & 1 \\ -\psi_1 & 0 & -\psi_2 & 1 & -\psi_3 & 0 \\ 0 & 0 & -\beta_{qp} & 0 & 1 & 0 \\ -(1-w_L) & 0 & 0 & w_S & -w_L & 1 \end{pmatrix} \begin{pmatrix} q_t^{pip} \\ ip_t \\ p_t \\ s_t \\ q_t^{LNG} \\ c_t \end{pmatrix} = \mathbf{B}x_{t-1} + \begin{pmatrix} u_t^{\text{pipeline supply}} \\ u_t^{\text{aggregate demand}} \\ u_t^{\text{flow demand}} \\ u_t^{\text{precautionary demand}} \\ u_t^{\text{LNG supply}} \\ u_t^{\text{balance residual}} \end{pmatrix}$$

where the structural shocks are assumed to be normally distributed, $u_t \sim N(0_{n \times 1}, \mathbf{D})$, and the contemporaneous parameters in $\mathbf{A}$ satisfy the sign restrictions listed in Table 1. These restrictions are implemented through the prior on $\mathbf{A}$ by assigning truncated distributions, which serves two purposes: it incorporates prior information about elements of $\mathbf{A}$ that the likelihood does not pin down, and it concentrates the posterior on values of $\mathbf{A}$ whose impact matrix $\mathbf{A}^{-1}$ has economically sensible signs. The rest of this subsection derives the six equations in turn.

We begin with the structural equations of the two supply components. Both pipeline and LNG imports are assumed to respond contemporaneously only to innovations in the gas price, with the sign restrictions $\gamma_{qp} > 0$ and $\beta_{qp} > 0$ and priors, described below, that keep both responses small. Because the production and delivery of natural gas involve adjustment costs, imports react to real activity shocks only with a delay of at least one month.

$$q_t^{pip} = \gamma_{qp}p_t + \mathbf{b}'_1x_{t-1} + \varepsilon_t^{\text{pipeline supply}} \quad (27)$$

$$q_t^{LNG} = \beta_{qp}p_t + \mathbf{b}'_5x_{t-1} + \varepsilon_t^{\text{LNG supply}} \quad (28)$$

Real activity is proxied by industrial production, which is assumed to respond contemporaneously only to the gas price. The reasoning is that a supply disruption reaches industrial activity through the price rather than through physical rationing, because in the short run

⁴March 2026 is the first full month of the Hormuz crisis, so the sample covers the first three months of that episode. The treatment of preliminary releases at the end of the sample is detailed in Appendix C.

firms can partially substitute other energy sources for gas. An increase in the gas price is therefore expected to reduce output, but modestly, so the corresponding elasticity $\alpha_{yp} < 0$ is assumed to be negative.

$$ip_t = \alpha_{yp} p_t + \mathbf{b}'_2 x_{t-1} + \varepsilon_t^{\text{aggregate demand}} \quad (29)$$

Demand for natural gas is separated into *flow demand* and *precautionary demand*. Flow demand is the demand for gas that is actually burned, and it is now written directly for observed consumption. Two contemporaneous channels enter: industrial production affects gas demand through the income elasticity $\alpha_{qy} > 0$, assumed to be positive, and the gas price affects it through the price elasticity $\alpha_{qp} < 0$, assumed to be negative.

$$c_t = \alpha_{qy} ip_t + \alpha_{qp} p_t + \mathbf{b}'_3 x_{t-1} + \varepsilon_t^{\text{flow demand}} \quad (30)$$

Precautionary demand captures changes in market participants' expectations that lead to adjustments in natural gas inventories. Such behaviour may arise either because agents anticipate a potential supply disruption or because a rise in prices triggers speculative motives among inventory holders. Stocks are adjusted in response to current imports, making inventories a potential substitute for them. Inventories also react to price innovations within the same month, because any shock that raises the gas price has two competing effects on stocks. On one hand, a higher price gives inventory holders the opportunity to realise a profit by selling above their acquisition cost, implying that stocks decline as the price rises ($\psi_2 < 0$). On the other hand, if inventory holders read the current price increase as a precursor of further increases, they may prefer to expand current inventories in order to sell more gas in the future ($\psi_2 > 0$). These two conflicting forces make the sign of ψ_2 theoretically ambiguous, and thus, like the other two inventory parameters ψ_1 and ψ_3 , it is left without a sign restriction.

$$s_t = \psi_1 q_t^{pip} + \psi_2 p_t + \psi_3 q_t^{LNG} + \mathbf{b}'_4 x_{t-1} + \varepsilon_t^{\text{precautionary demand}} \quad (31)$$

The sixth equation is the market balance. In levels, gas consumed in a month equals gas supplied minus the gas added to storage, so that in log growth rates the identity becomes a weighted sum of the growth rates of its components,

$$c_t = (1 - w_L) q_t^{pip} + w_L q_t^{LNG} - w_S s_t + \varepsilon_t^{\text{balance}}, \quad (32)$$

where w_L is the share of LNG in total supply and w_S converts a percentage change in the stock into the corresponding percentage of monthly consumption, that is, the average level of inventories expressed in months of consumption. Two features of this equation matter for what follows. First, w_L and w_S are not behavioural elasticities but aggregation weights: they answer the question of how much each component weighs inside a sum, not how any agent responds to a price. Second, the equation holds exactly in levels but only approximately in log growth rates, because the log-linearisation is taken around a point while the LNG share moved from roughly 15% to 45% within the sample. The approximation error, together with any measurement inconsistency between JODI trade flows, JODI stocks and Eurostat consumption, is carried by $\varepsilon_t^{\text{balance}}$. This term is therefore a measurement residual and not an economic shock, and its estimated importance serves as a check on the specification: were the identity a poor description of the data, this residual would absorb a large share of the variance of the price. In the estimates reported below, this residual accounts for less than 5% of the forecast error variance of the price of gas at any horizon.

Parameter	Meaning	Type	Sign restriction
γ_{qp}	Pipeline supply elasticity	elasticity	$\gamma_{qp} > 0$
β_{qp}	LNG supply elasticity	elasticity	$\beta_{qp} > 0$
α_{yp}	Effect of p on economic activity	elasticity	$\alpha_{yp} < 0$
α_{qy}	Income elasticity of gas demand	elasticity	$\alpha_{qy} > 0$
α_{qp}	Price elasticity of gas consumption	elasticity	$\alpha_{qp} < 0$
ψ_1	Effect of q^{pip} on gas inventories	elasticity	none
ψ_2	Effect of p on gas inventories	elasticity	none
ψ_3	Effect of q^{LNG} on gas inventories	elasticity	none
w_L	LNG share in total supply	aggregation weight	$0 < w_L < 1$
w_S	Inventories in months of consumption	aggregation weight	$w_S > 0$

Table 1: Contemporaneous parameters in **A**

4.3 Why observed consumption changes the identification

In the specification that is standard in this literature, the demand equation is written for a constructed quantity aggregate. Since consumers are indifferent to the source of the gas they burn, apparent consumption is defined as $Q_t = q_t^{pip} + q_t^{LNG} - s_t$ and the structural demand equation is imposed on that construction. This is a natural modelling choice and it is the one made in the closest antecedents of this paper. It has, however, a consequence for identification that has not been made explicit in the literature and that turns out to be decisive.

The variable q_t^{LNG} appears twice in the system: as the dependent variable of the LNG supply equation, with the price elasticity β_{qp} , and inside the left-hand side of the demand equation with a unit coefficient. Substituting the demand equation into the LNG supply equation, and recalling that every variable in the system, s_t included, is a monthly log change, the two rows deliver a single combination of the data,

$$q_t^{pip} - s_t = \alpha_{qy} ip_t + (\alpha_{qp} - \beta_{qp}) p_t + \mathbf{b}'x_{t-1} + \varepsilon_t^{\text{flow demand}} - \varepsilon_t^{\text{LNG supply}}, \quad (33)$$

in which the demand and LNG-supply elasticities enter only through their difference. The likelihood is therefore informative about $\alpha_{qp} - \beta_{qp}$ but carries very little information about how that difference is split between the two parameters. The two parameters are traded off along a posterior ridge, and where a given estimate sits on that ridge is decided by the priors rather than by the data. A tighter prior on demand does not resolve the problem; it simply pushes both parameters along the ridge together.

Replacing the constructed aggregate with observed consumption breaks the tie. The demand equation is then written for a variable that is measured independently of the two supply series,

$$c_t = \alpha_{qy} ip_t + \alpha_{qp} p_t + \mathbf{b}'_3 x_{t-1} + \varepsilon_t^{\text{flow demand}}, \quad (34)$$

so α_{qp} is now identified from the joint behaviour of consumption and the price, and β_{qp} from the joint behaviour of LNG arrivals and the price, with no shared variable forcing them onto a common ridge. In the estimates reported below the posterior correlation between the two is +0.20, and tightening or loosening the demand prior no longer moves the LNG-supply

elasticity. The accounting relationship that the constructed-aggregate specification imposed as an exact identity does not disappear; it re-enters as the balance equation introduced above, with its own error term whose magnitude serves as a diagnostic. The gain requires observed consumption to be an independent measurement rather than the identity recomputed: were consumption to satisfy the balance exactly, the sixth equation would be degenerate and would add no information. The small but nonzero balance residual reported in Section 5.1 confirms that it is such a measurement, with the identity holding well but not so exactly that the sixth equation carries nothing of its own.

4.4 Prior knowledge about these parameters

The structural parameters embedded in $\mathbf{A}$ can be disciplined using a well-established empirical literature on energy markets. In set-identified SVAR models that rely on inequality or sign restrictions, these parameters play a central role in shaping inference. Unlike in fully identified models, the posterior distribution does not converge to the likelihood even asymptotically, implying that prior information does not wash out with increasing sample size (Baumeister and Hamilton, 2015). The choice of prior distributions is therefore not innocuous and must be grounded in economically and empirically plausible values.

The short-run price elasticity of pipeline natural gas supply, γ_{qp} , measures the responsiveness of pipeline gas imports to price changes. We impose $\gamma_{qp} > 0$ and centre the prior at 0.1 with a scale of 0.2, reflecting the well-established view that natural gas supply is highly inelastic in the short run. Empirical studies consistently find small elasticities, typically close to zero or below 0.1, due to technological constraints, adjustment costs, and contractual rigidities (Al-Sahlawi, 1989; Dahl and Duggan, 1996; Arora, 2014; Rubaszek et al., 2021; Mason and Roberts, 2018). Centring the prior at a small positive value captures an upward-sloping but steep supply curve, while the heavy-tailed Student- t prior allows for uncertainty without admitting implausibly large elasticities.

The prior on the short-run price elasticity of LNG supply, β_{qp} , requires more care, because the mechanism it governs is not the same as the one that constrains pipeline gas. Our LNG variable measures net cargo receipts: LNG imported aboard ocean tankers, on a regasified-equivalent basis, net of re-exports. It therefore captures arrivals, not the volume regasified and sent out into the transmission grid in the same month. What responds to a European price increase within the month is consequently not liquefaction output, which is fixed on a monthly horizon by plant capacity, but the allocation of an already-produced and globally mobile stock of LNG: destination-flexible cargoes redirected toward Europe rather than Asia when the TTF-JKM spread widens, cargoes already in European waters discharged sooner rather than later, and re-exports from European terminals withheld. This reallocation channel is capable of more short-run responsiveness than production, and it is what the literature has in mind when it describes LNG as the flexible balancing source in Europe (Sharples, 2023; Ritz, 2019). The spread between the Asian and the European gas price that governs this reallocation has itself entered recent empirical work: Adolfsen et al. (2026) include it as an endogenous variable in a weekly model of the European market, precisely to let global LNG supply conditions reach the system through the price channel that redirects cargoes.

We impose $\beta_{qp} > 0$ and centre the prior at 0.1 with a scale of 0.2, the same conservative calibration used for pipeline supply. Anchoring both elasticities at the same near-rigid location means that any separation the model reports between γ_{qp} and β_{qp} is produced by the likelihood against the prior rather than assumed into it. Section 5.1 reports what the data deliver on this point.

On the demand side, the short-run price elasticity of gas consumption, α_{qp} , is expected to be negative. We centre the prior at -0.3 with a scale of 0.1 and truncate the distribution to enforce $\alpha_{qp} < 0$. This calibration reflects the broad consensus that gas demand is highly inelastic in the short run. Early evidence summarised by (Al-Sahlawi, 1989) documents short-run elasticities clustered around -0.3 , while more recent studies typically estimate elasticities between -0.04 and -0.16 (Joutz et al., 2009; Bernstein and Madlener, 2011; Arora, 2014), with demand remaining far from elastic even when allowing for heterogeneity (Burke and Yang, 2016). It is worth noting that in this specification the parameter is estimated against observed consumption rather than a constructed aggregate, so the posterior is informative about the price responsiveness of gas actually burned.

The elasticity of gas demand with respect to aggregate economic activity, α_{qy} , captures the income elasticity of natural gas demand. We impose $\alpha_{qy} > 0$ and centre the prior at 0.7 with a scale of 0.3, reflecting a strong but less than proportional response of gas consumption to output. This is motivated by survey evidence summarised by (Al-Sahlawi, 1989), which reports income elasticities ranging from near zero to above unity, and by cross-country estimates for OECD economies that cluster around 0.7 (Burke and Yang, 2016).

The parameter α_{yp} captures the contemporaneous response of aggregate economic activity to gas prices. We impose a negative sign restriction and centre the prior at -0.05 with a scale of 0.1, reflecting the view that higher energy prices act as a drag on output by raising production costs and reducing real disposable income, while having only a modest immediate effect (Kilian, 2008). Similar calibrations are standard in Bayesian SVAR studies of energy markets, and posterior estimates are typically close to zero (Baumeister and Hamilton, 2019). Evidence from the European gas crisis likewise points to noticeable but non-catastrophic short-run output effects (Güntner et al., 2024).

The three storage parameters ψ_1 , ψ_2 and ψ_3 are left unrestricted in sign, for the reasons given in the specification above, and their prior scales are set differently from those of the elasticities. Left agnostic, these couplings are resolved jointly with the supply elasticities along posterior ridges, so their treatment matters for the parameters of interest. The couplings ψ_1 and ψ_3 , however, are not behavioural responses to a price: they describe how storage comoves, within the month, with the volumes arriving through each import channel. That comovement is directly observable, because European storage stocks, external pipeline arrivals and terminal send-out are all published daily.⁵ A regression of monthly stock growth on the growth of both arrival aggregates, controlling for the price, heating demand and month-of-year effects, delivers coefficients of 0.015 (Newey–West standard error 0.065) for pipeline arrivals and 0.009 (0.035) for LNG send-out: within the month, storage does not

⁵By the European transparency platforms: Gas Infrastructure Europe for storage and LNG terminals, and ENTSOG for cross-border pipeline flows.

track arrivals. We therefore keep the prior centres at zero and set the scales to twice the measured standard errors, 0.13 for ψ_1 and 0.07 for ψ_3 , in the same spirit in which external information disciplines priors elsewhere in this class of models (Baumeister and Hamilton, 2019). The remaining coupling, ψ_2 , keeps the agnostic scale of 0.5, because the price is the one variable that the storage-demand shock itself moves within the month, so the analogous regression coefficient is not structural, whereas arrivals are largely fixed within the month by nomination schedules and shipping times. Appendix N presents the evidence in detail, including the figure behind these coefficients and the robustness of both.

The two aggregation weights require a different kind of prior, and we are explicit about why. Unlike the elasticities, w_L and w_S have directly observable counterparts: w_L is the average share of LNG in total supply and w_S is the average ratio of inventories to monthly consumption, both computable from the levels of the series without reference to the model.⁶ We centre their priors at those sample averages, 0.229 and 1.774 respectively, with scales of 0.1 and 0.5. Centring them at zero, as we do for the storage parameters, would amount to asserting a priori that LNG constitutes none of European supply and that Europe holds no gas in storage, statements that the data contradict directly. We note for transparency that centring a prior on a sample moment computed from the same dataset is a practice that deserves justification in a Bayesian setting. Two considerations apply here. First, the moments in question are averages of levels, whereas the likelihood of the SVAR is a function of growth rates; the information used to locate these priors is close to ancillary with respect to the dynamics that identify the remaining parameters. Second, the same practice is already standard in this class of models: the Minnesota prior on $\mathbf{B}$ calibrates its scales from residual variances of univariate autoregressions fitted to the same data, in this paper as in Baumeister and Hamilton (2019). An alternative specification that fixes w_L and w_S at their observed values without estimating them is available and delivers materially identical results.

In summary, all prior distributions are centred on values supported either by the empirical literature or, for the two aggregation weights and the storage couplings, by directly observable features of the data. Supply elasticities are small and positive, the demand price elasticity moderate and negative, the income elasticity positive and sizeable, the output response to gas prices modestly negative, and the storage couplings measured from daily data where a measured counterpart exists and agnostic where it does not. By adopting literature-based priors within the Bayesian framework of Baumeister and Hamilton (2015), the model achieves economically meaningful identification while avoiding improper posteriors.

5 Results

In this section we present the estimation results of the baseline six-variable model. We begin with the posterior distributions of the contemporaneous parameters, paying particular attention to the two supply elasticities and to the price elasticity of consumption, which

⁶The weight is a change of denominator. The balance identity needs the inventory flow expressed as a fraction of monthly consumption, $\Delta S_t/\bar{C}$, while the variable in the system is the growth of the stock, $s_t = \Delta \log S_t \approx \Delta S_t/\bar{S}$. The two share a numerator, so $\Delta S_t/\bar{C} = (\bar{S}/\bar{C})(\Delta S_t/\bar{S}) \approx w_S s_t$: the conversion factor is the average stock expressed in months of consumption.

Parameter	Interpretation	Prior	Posterior		
		mode (sd)	5%	median	95%
γ_{qp}	pipeline supply elasticity	0.10 (0.20)	+0.052	+0.116	+0.258
α_{yp}	activity response to price	-0.05 (0.10)	-0.040	-0.015	-0.002
α_{qy}	demand income elasticity	0.70 (0.30)	+0.406	+0.694	+1.013
α_{qp}	demand price elasticity	-0.30 (0.10)	-0.256	-0.098	-0.014
ψ_1	storage: pipeline supply	0.00 (0.01)	-1.205	-0.199	+0.071
ψ_2	storage: price	0.00 (0.50)	-0.880	-0.413	-0.231
ψ_3	storage: LNG supply	0.00 (0.01)	-0.108	+0.006	+0.086
β_{qp}	LNG supply elasticity	0.10 (0.20)	+0.032	+0.161	+0.343
w_L	LNG share in supply	0.23 (0.10)	+0.192	+0.228	+0.265
w_S	stock/consumption ratio	1.77 (0.50)	+1.614	+1.797	+1.990

Table 2: Priors and posteriors for the contemporaneous parameters in **A**. Prior locations and scales are those used in estimation; the posterior quantiles are computed from 300,000 retained draws of the baseline six-variable specification.

the specification of Section 4.3 identifies from separate moments for the first time. We then analyse the impulse responses, the forecast error variance decomposition and the historical decomposition of the real price of gas. We close with counterfactual exercises for the 2022–2023 crisis and with a joint analysis of the two European gas crises.

5.1 Posterior estimates of the elasticities

Table 2 reports prior locations and posterior quantiles for the ten contemporaneous parameters, and Figure 1 displays the corresponding prior and posterior densities. The estimates are based on 300,000 retained draws from a chain of 2 million, after 1.7 million burn-in draws, with an acceptance ratio of 0.32. The text below quotes posterior medians; credible intervals for every parameter are in the table.

The posterior median of the short-run price elasticity of pipeline-and-domestic supply, γ_{qp} , is 0.116: a 10% increase in the price of gas raises pipeline-and-domestic supply by little more than 1% within the month, so this channel is estimated to be steeply upward-sloping. The price elasticity of LNG supply, β_{qp} , has a posterior median of 0.160, comparable in magnitude. The posterior probability that LNG supply is more price-elastic than pipeline-and-domestic supply is 0.64: the data lean toward LNG being the more responsive of the two but do not establish the ordering in either direction.⁷

On the demand side, the posterior median of the short-run price elasticity of consumption,

⁷Three further checks support a small elasticity; Appendix O details them. The reduced-form correlation between the price and LNG-arrival innovations is 0.001; a regime-based estimator in the spirit of Rigobon (2003) returns a slope of 0.07 with a bootstrap 95% interval of $[-0.08, 0.20]$; and an elasticity of one would require a crisis-period covariance eighteen standard errors above the measured one.

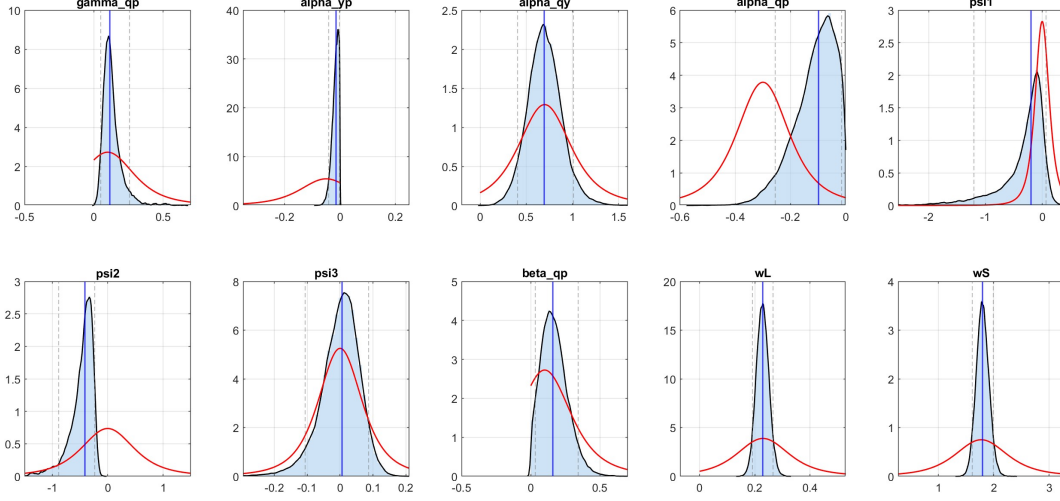


Figure 1: Prior and posterior densities of the ten contemporaneous parameters of $\mathbf{A}$ for the baseline six-variable model: the four elasticities of the supply, activity and consumption equations (γ_{qp} , α_{yp} , α_{qy} , α_{qp}), the three storage parameters (ψ_1 , ψ_2 , ψ_3), the LNG supply elasticity (β_{qp}), and the two aggregation weights of the balance equation (w_L , w_S). The light-blue filled histogram shows the marginal posterior; the solid black line is the corresponding kernel density; the vertical dashed grey lines mark the 5% and 95% posterior quantiles; the solid vertical blue line marks the posterior median; the solid red line is the prior density.

α_{qp} , is -0.098 . This is the first estimate in this framework that refers to observed gas consumption rather than to a constructed residual quantity, and it sits inside the range of existing short-run estimates: Joutz et al. (2009) report -0.09 for United States residential demand, Arora (2014) obtain values between -0.10 and -0.16 for aggregate United States demand, Bernstein and Madlener (2011) estimate an average of -0.24 across twelve OECD countries, and Burke and Yang (2016), whose focus is the long run, find household demand very inelastic at short horizons. The estimate is also economically coherent: at monthly frequency European gas consumption is overwhelmingly determined by weather and by the technical requirements of heating and industrial processes, leaving little room for a within-month price response.⁸

The income elasticity α_{qy} has a posterior median of 0.695 , close to the 0.727 of Rubaszek et al. (2021) and the 0.741 of Farag (2024) for the United States gas market, and to the 0.725 obtained by Baumeister and Hamilton (2019) for crude oil. The contemporaneous response of activity to the gas price, α_{yp} , is small at -0.015 , and estimates of this response are close to zero throughout the literature: -0.009 in Rubaszek et al. (2021), -0.005 in Farag (2024) and -0.002 in Baumeister and Hamilton (2019), while Güntner et al. (2024), whose identification does not estimate structural coefficients, likewise report only a moderate impact effect of gas prices on German industrial production.

The storage block is where the monthly likelihood is least informative on its own, and it is where the measured priors of Section 4.4 do their work. The posterior median of ψ_2 , the

⁸Consistent with that reading, the correlation between the innovation to the consumption equation and the heating-degree-day anomaly is 0.68 .

contemporaneous response of inventories to the price, is -0.413 , with a credible interval that lies entirely below zero: higher prices induce storage holders to release gas rather than to accumulate it. Storage, on these estimates, is the price-elastic side of the European gas market, not LNG arrivals. The two measured couplings behave as the daily evidence says they should. The parameter ψ_3 is centred on zero with a tight interval, and ψ_1 has a median of -0.198 , though its interval retains a long left tail where the likelihood pulls against the prior along the ridge; we do not interpret its location.

Finally, the two aggregation weights deliver the check announced in Section 4.2, and the logic of that check is simple: the balance equation is meant to be an accounting identity, so if the model describes the data correctly, the estimated weights should land on their physically observable values. They do. The posterior median of w_L is 0.228 against a sample LNG share of 0.229 , and that of w_S is 1.797 against an observed stock-to-consumption ratio of 1.774 .

5.2 Impulse Response Functions

We decompose the dynamics of the European natural gas market into six structural shocks: two supply shocks (pipeline-and-domestic, LNG), three demand shocks (aggregate demand, flow demand for consumption, precautionary or storage demand), and the balance residual, which is a measurement term and is not interpreted as an economic disturbance. Figure 2 traces the cumulative responses of the real price of gas, of consumption and of inventories to each of the five economic shocks; the full grid of responses of all six variables to all six shocks is reported in Appendix M.

Three regularities organise these responses, and they are the dynamic core of the paper. The two physical supply channels move the price slowly and almost permanently. The storage channel moves it immediately and transitorily. Consumption barely moves at all. The remainder of this subsection documents each in turn.

An adverse pipeline supply shock lowers pipeline imports on impact and raises the real price of gas, while industrial production declines modestly; inventories are drawn down, cushioning part of the physical shortfall, and consumption falls. The cumulative price response peaks at about five months and does not subsequently unwind, retaining 95% of its peak at two years. An adverse LNG supply shock removes cargoes from the European market and transmits in much the same way, with inventories drawn down as part of the missing volume is replaced from storage; its price response builds more slowly, peaking at about twelve months, and at two years its cumulative effect per standard deviation is comparable to that of the pipeline shock. In persistence the two physical supply shocks are therefore alike, and their effects on the price are close to permanent. Where they differ is in importance: the pipeline shock hits harder on impact and accounts for a larger share of the variance of the price in the decomposition of the next subsection.

Writing the demand equation for observed consumption separates the two demand shocks cleanly. A flow demand shock, an autonomous increase in the gas actually burned, driven predominantly by weather, raises the price and draws inventories down: part of the gas that is burned comes out of storage. A precautionary demand shock moves inventories the

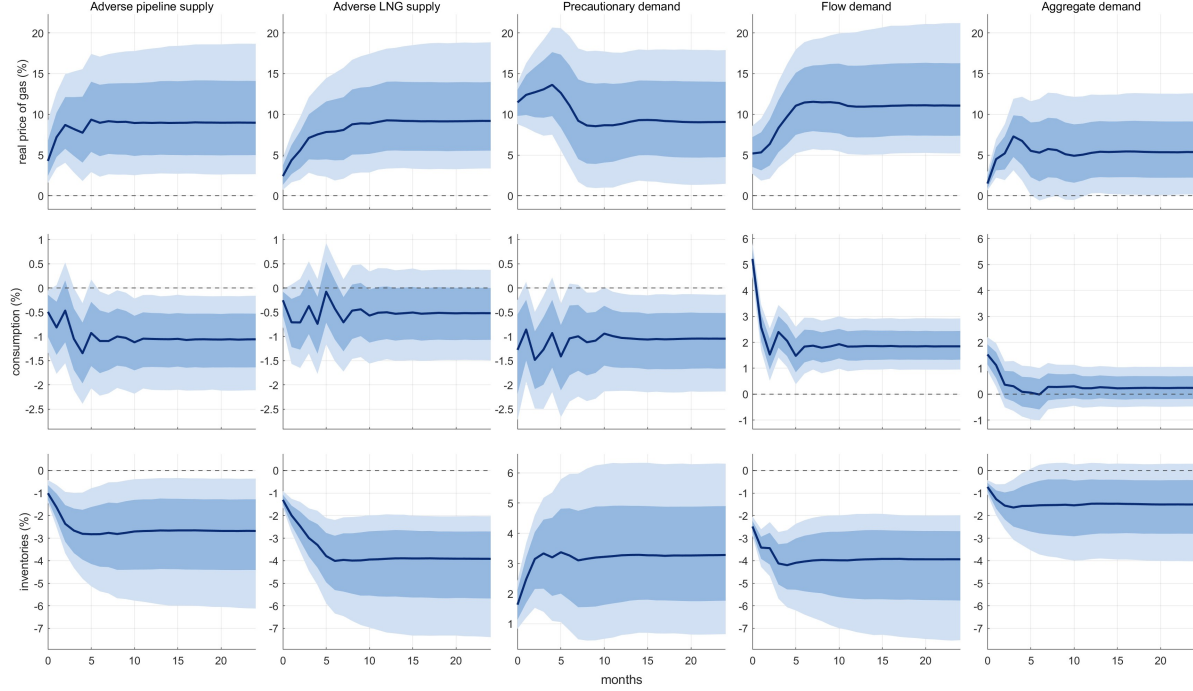


Figure 2: Cumulative responses of the real price of gas, of consumption and of inventories (rows) to the five economic structural shocks (columns); the balance residual, a measurement term, is omitted. The solid line is the posterior median and the shaded areas are the 68% and 90% credible bands; responses are in percent and the horizon is 24 months. The two supply shocks are shown as adverse, so every column reads as a disturbance from the European buyer’s standpoint. The full grid of responses of all six variables to all six shocks is in Appendix M.

opposite way, raising the price and inventories at the same time: gas is being bought to be stored, not burned. The sign of the price–inventory comovement is therefore what separates the two demand shocks. The impact effect of the precautionary shock on the price is the largest of any shock in the system. Its persistence is the feature that matters for the episodes analysed below: the response peaks within four months and then sheds a third of its peak by two years, a larger reversal than for any other shock, although the effect remains different from zero at every horizon. The expectational channel acts hard and then partially unwinds, while the physical channels build slowly and stay.

5.3 Variance Decomposition

Table 3 reports the median forecast error variance decomposition of the real gas price. The result is unambiguous and is the central quantitative finding of the paper: precautionary storage demand is the dominant source of price variation at every horizon. At twelve months, the horizon at which this class of decompositions is usually compared across studies and the one we treat as the headline, it accounts for 47% of the forecast error variance of the real price; flow demand and pipeline-and-domestic supply follow at 15% and 13%, with aggregate demand and LNG supply near 8% each. On impact the precautionary share is larger still, at 63%. That figure reflects the speed of the expectational channel rather than the composition

Shock	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 24$
Pipeline supply	8.4	12.3	13.0	13.0	13.0
Aggregate demand	1.1	5.3	7.8	8.2	8.3
Flow demand	12.8	12.2	14.4	14.5	14.6
Precautionary demand	62.6	53.0	47.4	46.7	46.5
LNG supply	2.9	5.4	6.8	7.5	7.6
Balance residual	4.2	4.8	4.7	4.6	4.7

Table 3: Median forecast error variance decomposition of the real gas price, in percent, at horizons of 1, 3, 6, 12 and 24 months. Rows do not sum to 100 because the table reports posterior medians rather than the shares of a single draw. Monthly data, 2014M2–2026M5.

of a typical month: an expectational disturbance is priced before any physical volume can move, so the one-month forecast error of the fastest-moving variable in the system loads on the shock that absorbs news.⁹ The split by subperiods reported below makes the distinction explicit.

The intuition for the dominance of storage is institutional. Within the month, consumption is fixed by weather and by the technical requirements of heating and industry, and both import channels are fixed by contracts, nomination schedules and shipping times. The one actor able to respond to news within the month is the holder of storage, so a revision of expectations about future supply or demand is expressed first, and mostly, as a change in the demand for stored gas. With every quantity nearly fixed in the short run, that change clears through the price. On impact, a one-standard-deviation precautionary shock raises the price by around 11% while moving no quantity by more than 2%: the stock build is financed by slightly lower consumption and slightly higher arrivals through both import channels, all of it price-induced. Storage demand carries the variance not because storage moves the most, but because it is the only quantity free to move at all.

A further feature of the decomposition is the position of LNG supply. LNG supply shocks account for 7.5% of the twelve-month forecast error variance. The volume of cargo arrivals, in other words, is not what moves the European gas price at monthly frequency, in line with the small posterior estimate of the LNG-supply elasticity and, as we show below, with what the historical decomposition attributes to each channel during both crisis episodes.

The attribution also stands at a distance from earlier estimates, and the two closest comparisons are instructive in different ways. Güntner et al. (2024) attribute 7.4% of the variance of the German gas price to storage demand, the smallest share among their four shocks, where we attribute 47% to it. Two differences account for most of the gap. Their price is the CPI-deflated German border import price, an average of prices set in long-term contracts,

⁹Anticipation of future supply raises the possibility of non-fundamentalness: if agents respond to news that current quantities do not yet reflect, a VAR in those quantities alone cannot recover the structural shocks. The standard remedy, due to Kilian and Murphy (2014), is to include forward-looking variables that price the news contemporaneously, which is what the spot price and inventories do in this system. Consistent with invertibility, the estimated precautionary shock is uncorrelated with both current and future pipeline and LNG supply shocks (Appendix A).

many of them indexed to oil or to lagged gas quotations, so a given piece of news enters that average only gradually over several months. The TTF spot price used here incorporates news in the month it arrives. Even at monthly frequency the distinction matters for the attribution: an expectational shock moves the spot price in the month of the news, whereas a contract-price average spreads the same news over many subsequent months, which shifts measured variance away from the impact of storage-demand shocks. Their specification, moreover, identifies demand without an observed consumption series; Section 4.3 sets out why that choice matters for how much price variation storage can be credited with.

The closest comparison of all is Adolfsen et al. (2026), who study the same market at weekly frequency with a different identification. Their historical decomposition attributes close to 40% of the changes in the TTF price over 2018–2024 to precautionary demand shocks, the dominant driver in their model as well, and they read the invasion month of February 2022 itself as a precautionary demand shock rather than as a supply shock. Two designs that share no identifying assumption, theirs built on weekly sign restrictions and the spread between the Asian and the European gas price, ours on monthly elasticity priors and observed consumption, arrive at the same ranking of the drivers of the price.

The United States studies that include inventories point the same way once the demand side is considered as a whole. Rubaszek et al. (2021) attribute 12–19% of the price variance to inventory demand but 65–79% to a market-specific demand shifter identified without an observed consumption series, and Farag (2024) attribute 28–32% to inventories and 46–54% to consumption demand, so in both cases the demand side as a whole carries on the order of three quarters of the variance. We find the same total: precautionary, flow and aggregate demand jointly account for roughly 70% at twelve months. Where this paper differs is the split within the demand side, and the difference is produced by the observed consumption series. A flow-demand shock disciplined by the gas actually burned cannot absorb expectational variance that has no counterpart in consumption, so the expectational component that a residual demand shifter would otherwise collect is attributed instead to storage demand, the channel through which it actually operates. The large precautionary share is, in this reading, the counterpart of measuring demand rather than inferring it.

The unconditional decomposition, finally, averages over regimes that the sample distinguishes sharply, and the average conceals what is arguably the more informative result. Table 4 examines whether that average is driven by the crisis windows. For each posterior draw, we compute the share of every shock in the sum of squared contributions to monthly price growth within a window, and the table reports the posterior median of each share.¹⁰ Over the full sample the realised shares reproduce the unconditional decomposition, with storage at 48%, and the calm years through 2020, computed the same way, deliver 41%, so the headline share is not a product of the two crises. What the windows add is that the dominance of storage intensifies during the two recent energy crises: the storage share rises to 63% over 2021–22 and to 87% over the three months of the 2026 episode, while the pipeline share falls

¹⁰The exercise is descriptive rather than exact: the contributions include the propagation of past shocks, and their cross-products are not attributed to any shock. Computing the shares within each draw, rather than from the posterior-median contribution paths, avoids understating shocks whose month-by-month attribution is uncertain.

Shock	Full sample (%)	2021–22 crisis (%)	2026 crisis (%)
Pipeline supply	13.0	9.6	2.3
Aggregate demand	8.5	2.1	0.9
Flow demand	13.5	11.0	2.9
Precautionary demand	48.1	62.6	87.0
LNG supply	7.8	3.8	1.2
Balance residual	4.1	3.4	2.7

Table 4: Regime split of the realised variance of the price contributions from the historical decomposition. Shares of the sum of squared monthly contributions within each window are computed draw by draw and reported at their posterior medians, in percent. The full sample is 2014M8–2026M5; the 2021–22 window covers September 2021 to December 2022 and the 2026 window March to May 2026. The 2026 window spans three months, and its shares should be read with that coarseness in mind.

from 14% in the calm years to 10% and then to 2%. Storage is the principal mover of the European gas price in every state of the market, and its dominance intensifies precisely when supply security is in question.

Taken together, the decomposition points to a market whose price is governed by expectations operating through storage in every state of the market, and never more so than under stress. That reading is what the historical decomposition and the counterfactual exercises examine next.

5.4 Historical Decomposition

A historical decomposition expresses the observed price path as the cumulative contribution of the identified structural shocks. Figure 3 presents it for the full sample, and Table 5 reports the attribution over the two crisis windows.

Before the decomposition is used to attribute the two crises, it is validated against episodes whose drivers are documented. The COVID-19 episode of 2020 appears as a demand-driven trough, with large negative contributions from flow and precautionary demand accompanying the collapse in activity and energy use (International Energy Agency, 2020). Prices then begin rising through 2021, well before the 2022 crisis proper: the decomposition attributes this to precautionary storage demand rather than to any contemporaneous supply loss, consistent with the combination of a strong Asian pull on globally traded cargoes (Reuters, 2021) and unusually low European storage entering the winter (Popkostova, 2022). The 2022 upheaval is dominated by pipeline supply and precautionary demand together (International Energy Agency, 2023), with aggregate demand contributing little, which matches contemporaneous assessments emphasising supply insecurity and risk premia over macroeconomic overheating (Furtuna et al., 2022). The normalisation through 2023 coincides with the unwinding of both forces, with strong storage buffers and the continued expansion of import capacity (European Commission, 2024; Institute for Energy Economics and Financial Analysis (IEEFA), 2024). The mild winter of 2022–2023 (Copernicus Climate Change Ser-

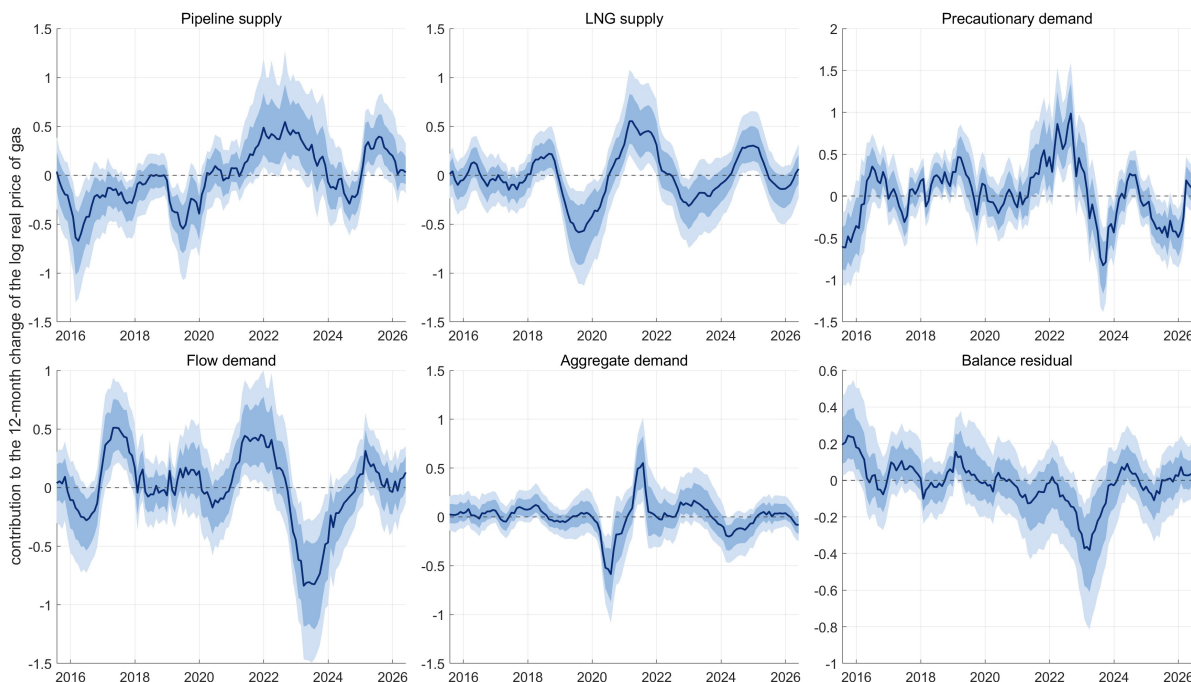


Figure 3: Historical decomposition of the European real price of gas, one panel per structural shock. Each panel shows the contribution of that shock to the change of the log price over the preceding twelve months: the solid line is the posterior median, the shaded areas the 68% and 90% credible bands. Separate panels make the uncertainty of each contribution visible, which a stacked decomposition cannot, and the twelve-month window keeps the accumulation of that uncertainty bounded. The balance-residual contribution is a measurement term and should not be read as an economic mechanism; its flatness is the graphical counterpart of the specification check reported in Section 5.1.

vice (C3S), 2023) and the autonomous saving of gas both enter the decomposition through the flow-demand channel. Separating them, which is possible because weather is exogenous, attributes roughly one third of the flow-demand effect on the price during the winter of the savings regulation to weather and two thirds to autonomous saving; Appendix G details the calculation, and the demand-side counterfactual of Section 5.5.3 should be read with that split in mind. The value the decomposition adds lies where the record is ambiguous, in the quantitative attribution of the two crises, to which we now turn.

The first crisis window runs from August 2021 to the August 2022 price peak and covers the run-up to and aftermath of the Russian invasion of Ukraine: record-low storage entering the winter of 2021, the invasion in February 2022, and the staged interruption of the Nord Stream flows through the summer. Over this window the observed real price of gas rose by a factor of 4.16. Precautionary storage demand contributed 70% of that rise and pipeline-and-domestic supply shocks 37%, while the contribution of LNG supply is negative, at -10% , so LNG acted to restrain the rise rather than to drive it (Table 5). The three channels jointly account for 96% of the observed increase, with the balance residual and the two remaining demand shocks accounting for the small remainder. The decomposition apportions the observed move among the shocks, including the propagation of shocks dated before the

Shock	Share of the observed move (%)	
	2021–22 crisis (Aug-21 → Aug-22)	2026 Hormuz (Feb-26 → May-26)
Pipeline supply	37	8
Aggregate demand	4	−16
Flow demand	4	10
Precautionary demand	70	54
LNG supply	−10	21
Balance residual	−10	20

Table 5: Attribution of the observed price move over each crisis window, from the historical decomposition, in percent of the reconstructed move (posterior medians). Signed shares can exceed 100% or turn negative when shocks offset. The observed moves are a factor of 4.16 for the 2021–22 window (+1.425 in logs) and a factor of 1.41 for the 2026 window (+0.340 in logs).

window opened; the complementary question, what the price would have been had the crisis shocks not occurred, is answered in Section 5.6. The conclusion from the attribution is that storage holders bidding for gas ahead of winter carried the larger part of the rise to the peak, on top of a physical loss of pipeline volumes whose contribution is smaller at the peak partly for a mechanical reason: the impulse responses of Section 5.1 show that the price response to a pipeline shock takes about five months to build, so much of the effect of the cuts of that spring and summer had not yet arrived by August 2022. The same responses imply that the pipeline loss is the persistent component of the two, while LNG acted as a partial buffer rather than as a source of the shock.

The 2026 Hormuz episode, covering February to May 2026, is a smaller event driven by different forces: the observed price rose by a factor of 1.41. Precautionary storage demand contributes 54% of the rise, LNG supply 21%, flow demand 10% and pipeline supply 8%. How these contributions are distributed across the months of the episode carries the more interesting information, and Section 5.6 examines it.

5.5 Counterfactual Analysis

The identified model can answer three what-if questions about the 2022–2023 crisis. The first freezes LNG imports at their pre-invasion level, removing the channel that in fact replaced the missing Russian gas. The second lets LNG absorb the pipeline shortfall in full, as if regasification terminals and global cargo supply had posed no limit. The third removes the autonomous reduction in gas demand over the winter of the European savings regulation, the gas that households, firms and governments chose not to burn. The first two exercises impose a path on LNG imports over a window, with the LNG-supply innovation absorbing the constraint; the third sets the flow-demand innovation itself to zero over the window of the regulation. In every case all other innovations remain at their historical values, as in Güntner et al. (2024) and Antolín-Díaz et al. (2021). Table 6 summarises the headline results; Figure 4 reports the three exercises side by side, and the imposed paths are in the appendices.

Scenario	Description	Terminal gap (%)		
		price	consumption	inventories
CF1	LNG frozen at its Feb-2022 level	+23.7	-1.2	-8.6
CF1b	LNG grows at its pre-crisis trend	-13.2	+0.8	+6.1
CF2	LNG fully absorbs the pipeline shortfall	-19.0	+1.1	+9.6
CF3	No autonomous demand reduction	+95.5	+11.5	-20.9

Table 6: Summary of the counterfactual scenarios. Terminal gaps are posterior medians of the counterfactual path relative to the observed one at the end of the sample, in percent. The response of observed consumption is a model output rather than an accounting residual.

The three exercises answer a common policy question, namely which of the adjustment channels available to Europe in 2022 mattered most for the price, and the terminal gaps of Table 6 rank those channels. Demand conservation is the largest single channel by a wide multiple: without it the price would have ended the sample about 95% higher, whereas the two LNG exercises, which bracket the LNG channel between complete unavailability and perfect elasticity, move the terminal price by +24% and −19% respectively. The policy implication is that the reduction of demand and the drawdown of storage contributed more to containing the 2022 crisis than the expansion of LNG receipts did. The subsections that follow describe each exercise in turn.

5.5.1 Counterfactual 1: LNG frozen at its February-2022 level

Russia invaded Ukraine in February 2022, and its pipeline deliveries to Europe collapsed over the following two years; the gas that replaced them arrived mostly as LNG. The first exercise asks what the crisis would have looked like without that channel: it holds net LNG imports at their February 2022 level until December 2023, so that the flexible channel is unavailable during the collapse in Russian pipeline deliveries. The real price of gas ends the sample about 24% higher than observed (Figure 4, left column). Inventories are 9.0% lower, and observed consumption is essentially unchanged at −1.2%: with LNG frozen, it is the drawdown of inventories that makes up for the missing Russian gas.

Where the adjustment falls is as informative as the size of the price effect. That consumption barely moves is a direct consequence of its near-zero price elasticity: European consumers, on these estimates, would have burned nearly the same volume of gas at prices a quarter higher, and the shortfall is met by storage instead. Appendix E reports the imposed path. Counterfactual 1b, summarised in Table 6 and detailed in Appendix I, is a less extreme variant in which LNG grows along its pre-crisis trend of 1.28% per month rather than being frozen; the price then ends about 13% below the observed path, indicating that even the secular expansion of LNG capacity, absent the crisis surge, would have delivered some relief.

5.5.2 Counterfactual 2: LNG fully absorbs the pipeline shortfall

The converse exercise imposes an LNG path that rises one-for-one with the contemporaneous shortfall in pipeline-and-domestic supply relative to February 2022, so that LNG supply is

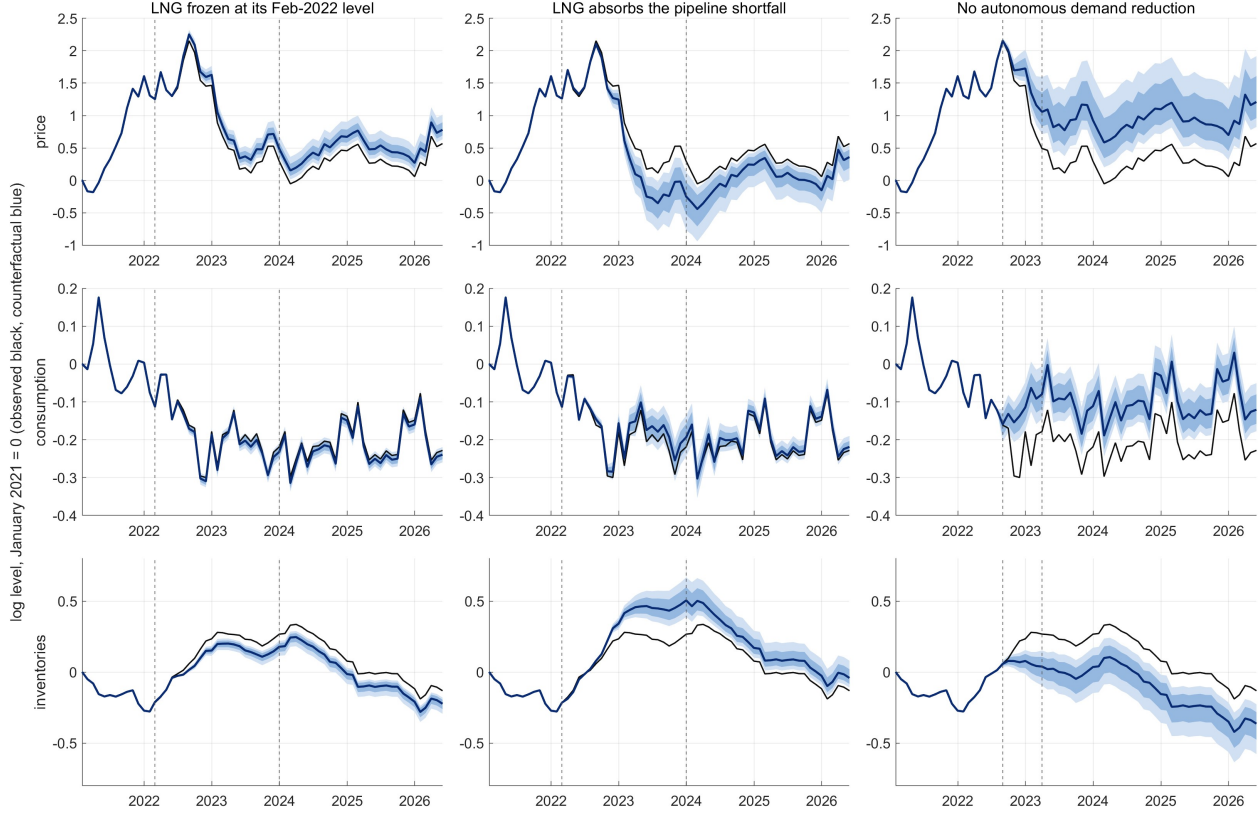


Figure 4: The three counterfactuals. Each column is one exercise: LNG imports frozen at their February 2022 level (left), LNG imports absorbing the pipeline-and-domestic shortfall in full (centre), and no autonomous demand reduction over the window of Council Regulation 2022/1369 (right). Each row shows the log level of one variable, normalised to zero in January 2021: the observed path in black and the counterfactual median in blue with 68% and 90% credible bands. Vertical dashed lines mark each constraint window; before it, the counterfactual coincides with the observed path by construction. Rows share a common scale across columns. Appendix D reports the same exercises as gaps from the observed path.

perfectly elastic. The real price of gas ends about 19% below the observed path, with inventories 9.2% higher (Figure 4, centre column). At first sight, replacing one source of gas with another might be expected to leave the price unchanged. The reason it falls instead is that the observed replacement was neither complete nor free: LNG covered only part of the missing pipeline gas, and the cargoes came because European prices rose far enough to pull them away from Asian buyers. The counterfactual delivers every missing unit without the price having to do that work, so the scarcity that produced the observed spike never arises. The imposed path requires a near-doubling of monthly LNG imports at the worst of the crisis (Appendix H), volumes well beyond the regasification, downstream-pipeline and global-cargo capacity available in 2022–2023.

Read jointly with Counterfactual 1, the two exercises bracket the contribution of the LNG channel: closing it entirely would have raised the terminal price by roughly 24%, and making it perfectly elastic would have lowered it by roughly 19%. Both magnitudes are modest, and the reason is the small β_{qp} : the LNG channel operates through that elasticity, and a channel

that responds little to the price also moves the price little when its path is imposed.

5.5.3 Counterfactual 3: no autonomous demand reduction

In August 2022, at the height of the crisis, the Council of the European Union adopted Regulation 2022/1369, committing member states to reduce their gas demand by 15% over the following winter. Over the window of the Regulation, August 2022 to March 2023, the flow-demand innovation, which takes systematically negative values as demand falls, is set to zero, so that the autonomous reduction in European gas demand never happens. The real price of gas ends the sample about 95% higher than observed. Observed consumption ends 10.9% higher and inventories 23.4% lower (Figure 4, right column).

Because the demand equation is estimated on observed consumption, the shock switched off is an autonomous movement in the gas actually burned, and the exercise measures what it claims to measure. The consumption figure is a direct estimate of the persistence of the demand reduction: some 11% of consumption at the end of the sample is attributable to the autonomous adjustment that began in the policy window.

One caveat is that the flow-demand innovation bundles three forces that lowered gas demand at the same time: the Regulation, the mild winter of 2022–2023, and voluntary conservation by households and firms. The weather part can be separated, because temperature is exogenous and measurable: Appendix G projects the flow-demand shock on a heating-degree-day anomaly and attributes roughly one third of the window’s flow-demand effect on the price to the mild winter, leaving two thirds that was genuinely autonomous saving. Within that autonomous part, the Regulation cannot be told apart from voluntary conservation, because both reduce the gas burned in the same way. Independent accounts of how European gas use was reshaped over this period reach the same conclusion about the joint nature of the adjustment (Zhou et al., 2025). Appendix F displays the shock being switched off and Appendix J extends the window to December 2023.

5.6 A joint anatomy of the two crises

Europe’s second gas crisis in five years arrived on 28 February 2026, when the closure of the Strait of Hormuz cut the route through which Qatari cargoes, and with them a substantial share of globally traded LNG, reach the market. The closure threatened the channel on which the previous crisis had left Europe dependent: the price rose 45% in March, before arrivals had fallen at all, and the truce of 7 April set the decline in motion. Because the episode falls inside the estimation sample, both crises can be examined with the same identified model, and the purpose of this subsection is to establish whether they worked through the same channels, and in what proportions. To that end we delete, over each crisis window, the structural innovations of the three channels through which a supply-security event can plausibly operate, pipeline supply, precautionary storage demand and LNG supply, one at a time and jointly, and we re-simulate the price path with all other innovations at their historical values; the gap between the observed and the re-simulated path measures what each channel contributed. The 2021–2022 window opens in September 2021 rather than at the invasion, because the crisis did: storage entered the winter of 2021 at record lows and

the price was already climbing months before February 2022, a run-up that the historical decomposition attributes to precautionary demand. Ending the window in December 2022 covers the peak and the first phase of the normalisation.

The results appear in Figure 5. The 2021–2022 crisis is led by storage demand and sustained by the pipeline loss: without the precautionary shocks of the window the August 2022 peak would have been 65% lower, without the pipeline shocks 38% lower, and without the favourable LNG shocks of the period roughly 28% higher. These figures need not coincide with the historical decomposition of Table 5, which also counts the propagation of shocks dated before the window; in practice the two agree closely for the pipeline and precautionary channels and differ only for LNG, whose adverse shocks are dated in 2021, before the window opens. For 2021–2022 the counterfactual thus largely confirms what the decomposition already established, and its role here is to put the two crises on the same footing. The new information concerns 2026, which the same exercise shows to be a different kind of crisis, dominated by the expectational channel: without the precautionary shocks the March 2026 peak would have been about 30% lower, against less than one percent for LNG supply. Contemporaneous assessments of the closure emphasised the exposure of Qatari volumes and the concentration of global LNG traffic through the strait (International Energy Agency, 2026; Kiel Institute for the World Economy, 2026). In the model this appears as storage demand: Europe bid for gas to hold against a Qatari shortfall that had not yet arrived, and the price rose before any cargo was missed. The extra stocks came from both sides of the balance, and the impulse responses say which: after a precautionary shock, consumption falls and arrivals rise on impact. March 2026 delivered exactly that combination, consumption down 7.9% and LNG arrivals up 3.3%.

The 2026 price peaked in March, the first month of the window, when European LNG arrivals had in fact risen by 3.3% relative to February; the physical shortfall arrived in April, when arrivals fell by 14.9%, of which the model attributes -7.4 percentage points to the LNG-supply shock itself and -4.8 to the unwinding of precautionary demand, which had been pulling cargoes toward Europe and released them as the price fell after the 7 April truce. A channel whose shock has not yet occurred cannot contribute to a price peak, so the near-zero March figure for LNG measures chronology rather than irrelevance. By the end of the window, removing the LNG shocks would lower the price by about 4%, while the effect of removing the precautionary shocks has decayed from 30% to 11%.

This asymmetry of timing, an expectational channel that acts immediately and decays and a physical channel that arrives with a shipping lag and accumulates, is the mechanism the model recovers from the 2026 data, and it is the same signature the impulse responses carry, with the price response to a precautionary shock peaking on impact and the response to a supply shock building over months. The asymmetry explains an otherwise puzzling feature of the episode: the European price rose 45% in the month when more gas arrived and fell 16% in the month when much less did. Prices move at the speed of information, cargoes at the speed of shipping. We note that a supply-security event whose immediate expression is a shift in storage demand will be recorded by this model as a precautionary shock rather than as a supply shock; identifying the announcement component as a supply-news shock would require a forward-looking variable and an event-based identification in the manner

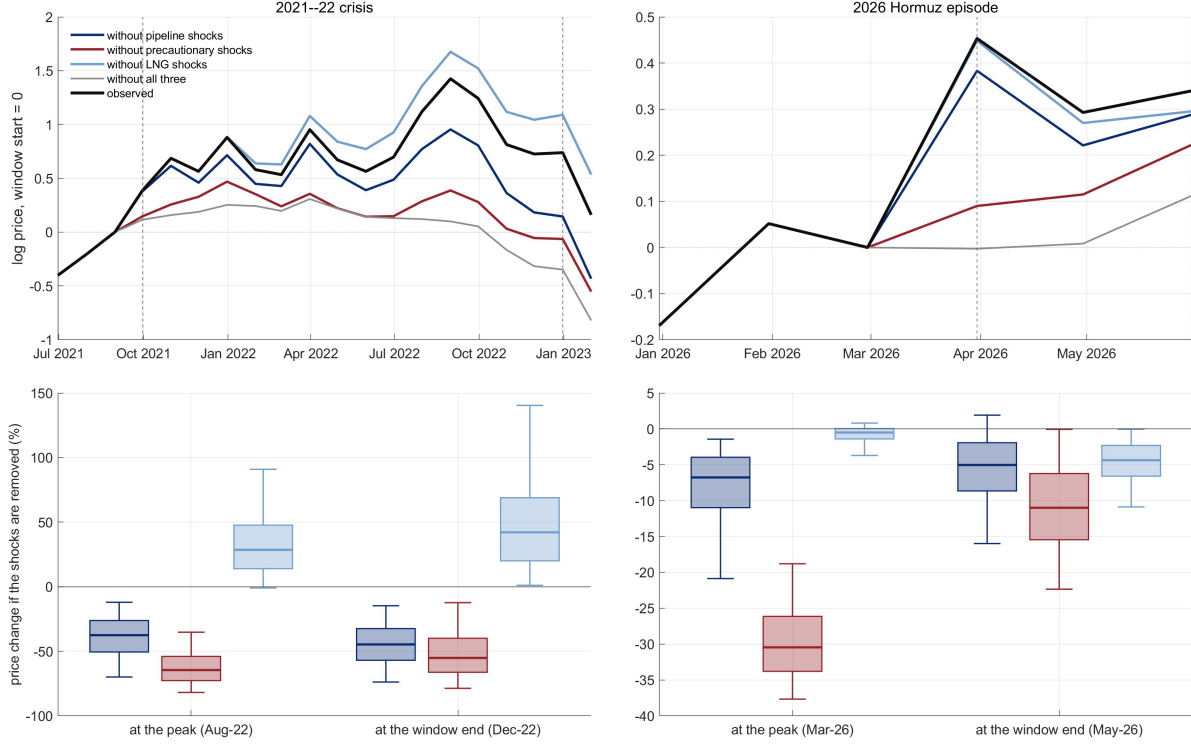


Figure 5: Anatomy of the two crises. *Top row*: the log real price, normalised to zero in the month before each window opens (black), against the counterfactual paths obtained by deleting, over the crisis window, the pipeline-supply shocks (dark blue), the precautionary-demand shocks (dark red), the LNG-supply shocks (light blue) and the three jointly (grey). *Bottom row*: the percentage change of the price at the peak and at the end of the window if the channel’s shocks are removed, the magnitudes quoted in the text, summarised as box-and-whisker plots of the posterior: the box spans the interquartile range, the line inside it is the median, and the whiskers reach the 5th and 95th percentiles. Each crisis is on its own scale. The 2026 peak is the first month of its window, so the physical LNG shortfall has not yet arrived there; see the discussion in the text. Appendix K reports each counterfactual path with its credible bands.

of Känzig (2021), a route that Alessandri and Gazzani (2025) have begun to explore for European gas with a news-based instrument and that we leave for future work within the present framework.

For completeness, Appendix L reports a forward-looking scenario exercise for the remainder of the Hormuz episode, in which paths for the LNG-supply and precautionary innovations are imposed over a twelve-month projection horizon. That exercise is illustrative rather than inferential, its scenario paths are calibrated by us rather than recovered from data, but it makes the mechanism quantitatively visible: under a winter-escalation scenario, 94% of the peak price gap is attributable to the expectational channel and approximately 4% to the volume channel.

The robustness checks that support these estimates, from convergence and prior sensitivity to the alternative IEA supply data, are collected in Appendix A.

6 Conclusions

This paper has studied the European natural gas market through a Bayesian structural vector autoregression that separates pipeline gas from LNG and writes the demand equation for observed gas consumption. The gain lies in the separation of the two supplies. Their price elasticities turn out to be similar, both small, with LNG somewhat the more elastic, but their roles during the recent crises were not the same. A model that lumps them into one aggregate sees only the total, and the composition of supply, where much of the adjustment of the last five years took place, drops out of the analysis.

The first substantive finding is that neither import channel responds much to the price within the month. Both supply elasticities are small, and the data lean toward LNG being the more responsive of the two without settling the ordering. The flexibility of LNG is real, but it lives at the horizon of cargo reallocation rather than inside the month, where arrivals are almost as predetermined as pipeline flows.

What the data identify sharply is storage. Inventories respond negatively to the price while consumption barely responds at all, governed as it is by weather and by the technical requirements of heating and industry, and precautionary storage demand is the largest single source of price variation at every horizon, accounting for close to half of the variance of the real price at one year. That dominance is not a product of the two crises: it is present in the calm years and it intensifies, rather than appears, when supply security comes into question. The European gas price, on these estimates, is set by expectations operating through storage rather than by the contemporaneous composition of import flows.

The counterfactual exercises for the 2022–2023 crisis order the policy channels accordingly. The autonomous reduction in gas demand did roughly three times as much to contain the price as either LNG scenario, whether the LNG channel is shut down entirely or made perfectly elastic, and with LNG frozen European consumption would have been almost unchanged, the shortfall coming out of storage. The identified reduction bundles the effect of policy, a mild winter and voluntary conservation; the weather part is measurable, at roughly one third, while policy and voluntary conservation remain bundled in the rest.

Because the 2026 episode falls inside the estimation sample, the two European gas crises can be read in a single frame. The 2021–2022 crisis is led by storage demand and sustained by a pipeline loss that is smaller at the peak but far more persistent, with LNG acting as a buffer rather than a source of the shock. The 2026 Hormuz episode is expectational almost in its entirety, and its timing is the instructive part: the price peaked in the month the news arrived, when cargoes had in fact risen, and fell in the month the physical shortfall landed. The physical component of a chokepoint disruption reaches Europe with a shipping lag while its expectational component arrives with the news, and a model that separates storage demand from arrivals can attribute each month to the channel that was actually operating.

One limitation matters for how the results should be read, and it concerns the labelling of the expectational channel. A supply-security event whose first expression is a rush for storable gas is recorded here as precautionary demand, the same convention the oil literature

uses, so the model cannot call the announcement component of a disruption a supply event. Doing so would require a forward-looking variable and an event-based identification in the manner of Känzig (2021), a route that Alessandri and Gazzani (2025) have begun to explore for European gas and that we regard as the most promising extension of the present paper.

Other avenues suggest themselves. Modelling international spillovers through the global LNG market, embedding the identified shocks in a macro-financial model, or disaggregating demand by sector would each sharpen the analysis of how energy disturbances transmit to inflation, output and asset prices. As JODI and Eurostat coverage of 2026 completes, the retrospective anatomy of the Hormuz episode reported here will extend automatically and its precision will improve.

The policy reading is more specific than the one that motivated this project. Europe's response to the loss of Russian pipeline gas rested on two policies: supply diversification, replacing pipeline deliveries with LNG from the international market, and the saving of gas. Both mattered, but not equally, with the LNG that arrived softening the 2022 peak and the saving of gas containing the price by more still. The 2026 episode was different, driven by expectations rather than by lost volumes, and it shows the cost of leaving them unanchored, since the price peaked in the month the strait closed, before any cargo had been missed, and most of that rise was precautionary bidding for storage. Because supply-security concerns are priced through storage, the instruments that complete the list are those acting on that channel: storage capacity and its filling trajectory, the credibility of the rules governing it, and the management of expectations during a disruption, together with the receiving-side infrastructure that determines which physical adjustments are available at short notice.

References

- Adolfson, J. F., Ferrari Minesso, M., Mork, J. E., and Van Robays, I. (2024). Gas price shocks and euro area inflation. *Journal of International Money and Finance*, 149:103183.
- Adolfson, J. F., Lappe, M.-S., Manu, A.-S., Rößler, D., Schupp, F., and Stalla-Bourdillon, A. (2026). Gas market shocks: Tracing the effect on euro area inflation expectations. Working Paper Series 3227, European Central Bank.
- Al-Sahlawi, M. A. (1989). The demand for natural gas: A survey of price and income elasticities. *The Energy Journal*, 10(1):77–90.
- Alessandri, P. and Gazzani, A. (2025). Natural gas and the macroeconomy: Not all energy shocks are alike. *Journal of Monetary Economics*, 151:103749.
- Alquist, R. and Kilian, L. (2010). What do we learn from the price of crude oil futures? *Journal of Applied Econometrics*, 25(4):539–573.
- Antolín-Díaz, J., Petrella, I., and Rubio-Ramírez, J. F. (2021). Structural scenario analysis with svars. *Journal of Monetary Economics*, 117(C):798–815.
- Arora, V. (2014). Estimates of the price elasticities of natural gas supply and demand in the united states. MPRA Paper 54232, University Library of Munich, Germany.

- Baumeister, C. and Hamilton, J. D. (2015). Sign restrictions in bayesian structural vars. *Journal of Business & Economic Statistics*, 33(1):1–13.
- Baumeister, C. and Hamilton, J. D. (2019). Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks. *American Economic Review*, 109(5):1873–1910.
- Bernstein, R. and Madlener, R. (2011). Residential natural gas demand elasticities in OECD countries: An ARDL bounds testing approach. FCN Working Paper 15/2011, Institute for Future Energy Consumer Needs and Behavior (FCN), RWTH Aachen University.
- Boeck, M. and Zörner, T. O. (2025). Natural gas prices, inflation expectations, and the pass-through to euro area inflation. *Energy Economics*, 141:108061.
- Burke, P. J. and Yang, H. (2016). The price and income elasticities of natural gas demand: International evidence. *Energy Economics*, 59:466–474.
- Casoli, C., Manera, M., and Valenti, D. (2024). Energy shocks in the Euro area: Disentangling the pass-through from oil and gas prices to inflation. *Journal of International Money and Finance*, 147:103154.
- Copernicus Climate Change Service (C3S) (2023). European state of the climate 2022 – annual temperature and winter anomalies. <https://climate.copernicus.eu/esotc/2022>. Accessed May 2026.
- Dahl, C. and Duggan, T. E. (1996). U.s. energy product supply elasticities: A survey and application to the u.s. oil market. *Resource and Energy Economics*, 18(3):243–263.
- Deaton, A. and Laroque, G. (1992). On the behaviour of commodity prices. *The Review of Economic Studies*, 59(1):1–23.
- European Commission (2024). Quarterly reports confirm significant recovery on eu gas and electricity markets in 4th quarter 2023. European Commission, Directorate-General for Energy.
- Farag, M. (2024). Revisiting the dynamics and elasticities of the u.s. natural gas market. EWI Working Papers 2024-8, Energiewirtschaftliches Institut an der Universitaet zu Koeln (EWI).
- Furtuna, O., Grassi, A., Ianiro, A., Kallage, K., Koci, R., Lenoci, F., Sowiński, A., and Vacirca, F. (2022). Financial stability risks from energy derivatives markets. Financial stability review, special feature, European Central Bank.
- Güntner, J., Reif, M., and Wolters, M. (2024). Sudden stop: Supply and demand shocks in the german natural gas market. *Journal of Applied Econometrics*, 39(7):1282–1300.
- Hamilton, J. D. (1996). This is what happened to the oil price-macroeconomy relationship. *Journal of Monetary Economics*, 38(2):215–220.

Hamilton, J. D. (2023). Comment on “the economic effects of a stop of energy imports from russia”. *Brookings Papers on Economic Activity*, Fall:456–481.

Institute for Energy Economics and Financial Analysis (IEEFA) (2024). Two years of REPowerEU: Gas reduction on track but latest target raises risks. <https://ieefa.org/resources/two-years-repowereu-gas-reduction-track-latest-target-raises-risks>. Accessed May 2026.

International Energy Agency (2002). Flexibility in natural gas supply and demand.

International Energy Agency (2020). Gas 2020. Technical report, International Energy Agency (IEA). Published 10 June 2020.

International Energy Agency (2023). Gas market lessons from the 2022-2023 energy crisis. Technical report, International Energy Agency (IEA). Report page.

International Energy Agency (2026). Middle east crisis disrupts international natural gas markets and delays global LNG supply wave. <https://www.iea.org/news/middle-east-crisis-disrupts-international-natural-gas-markets-and-delays-global-lng-s> Accessed May 2026.

Joutz, F. L., Shin, D., McDowell, B., and Trost, R. P. (2009). Estimating regional short-run and long-run price elasticities of residential natural gas demand in the u.s. USAEE Working Paper 09-021, United States Association for Energy Economics. Presented at the 28th USAEE/IAEE North American Conference, New Orleans, 2008. Available at SSRN.

Känzig, D. R. (2021). The macroeconomic effects of oil supply news: Evidence from OPEC announcements. *American Economic Review*, 111(4):1092–1125.

Kiel Institute for the World Economy (2026). The cost of closing the strait of Hormuz. https://www.kielinstitut.de/fileadmin/Dateiverwaltung/IfW-Publications/fis-import/01b7c020-27e6-4096-8cc5-e037738d2058-KPB_206.pdf. Accessed May 2026.

Kilian, L. (2008). The economic effects of energy price shocks. *Journal of Economic Literature*, 46(4):871–909.

Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review*, 99(3):1053–69.

Kilian, L. and Lee, T. K. (2014). Quantifying the speculative component in the real price of oil: The role of global oil inventories. *Journal of International Money and Finance*, 42:71–87.

Kilian, L. and Murphy, D. P. (2012). Why agnostic sign restrictions are not enough: Understanding the dynamics of oil market var models. *Journal of the European Economic Association*, 10(5):1166–1188.

- Kilian, L. and Murphy, D. P. (2014). The role of inventories and speculative trading in the global market for crude oil. *Journal of Applied Econometrics*, 29(3):454–478.
- López, L., Odendahl, F., Párraga Rodríguez, S., and Silgado-Gómez, E. (2024). The pass-through to inflation of gas price shocks. Working Paper Series 2968, European Central Bank.
- Lütkepohl, H. and Kilian, L. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press.
- Mason, C. F. and Roberts, G. (2018). Price elasticity of supply and productivity: An analysis of natural gas wells in wyoming. *The Energy Journal*, 39(1_suppl):79–100.
- Moll, B., Schularick, M., and Zachmann, G. (2023). The power of substitution: The great german gas debate in retrospect. *Brookings Papers on Economic Activity*, 2023(2):395–481.
- Molnar, G. (2022). Economics of gas transportation by pipeline and lng. In *The Palgrave Handbook of International Energy Economics*, volume None of *Springer Books*, pages 23–57. Springer.
- Oxford Institute for Energy Studies (2022). International gas contracts. Technical report, Oxford Institute for Energy Studies.
- Pindyck, R. S. (2001). The dynamics of commodity spot and futures markets: A primer. *The Energy Journal*, 22(3):1–29.
- Popkostova, Y. (2022). Europe’s energy crisis conundrum: Origins, impacts and way forward. EUISS Brief 2, European Union Institute for Security Studies (EUISS).
- Reuters (2021). On the cusp of europe’s winter season, gas storage hits 10-year low. Reuters.
- Rigobon, R. (2003). Identification through heteroskedasticity. *The Review of Economics and Statistics*, 85(4):777–792.
- Ritz, R. A. (2019). A strategic perspective on competition between pipeline gas and lng. *The Energy Journal*, 0(Number 5):None.
- Rubaszek, M., Szafranek, K., and Uddin, G. S. (2021). The dynamics and elasticities on the u.s. natural gas market. a bayesian structural var analysis. *Energy Economics*, 103:105526.
- Sedlar, D. K. et al. (2023). Influence of lng supply on gas market flexibility: Croatia case study. *Utilities Policy*.
- Sharples, J. (2023). A brave new world? lng contracts in the context of market turbulence and an uncertain future. OIES Paper NG 187, Oxford Institute for Energy Studies.
- Szafranek, K. and Rubaszek, M. (2024). Have european natural gas prices decoupled from crude oil prices? evidence from tvp-var analysis. *Studies in Nonlinear Dynamics & Econometrics*, 28(3):507–530.

- Ulvestad, M. et al. (2012). Natural gas and co2 price variation: impact on the relative ...
Energy Policy.
- Working, H. (1949). The theory of price of storage. *American Economic Review*, 39(6):1254–1262.
- Zhou, C., Zhu, B., Halff, A., Davis, S. J., Liu, Z., Bowring, S., Ben Arous, S., and Ciais, P. (2025). Europe’s adaptation to the energy crisis: reshaped gas supply–transmission–consumption structures and driving factors from 2022 to 2024. *Earth System Science Data*, 17:3431–3446.

A Robustness

This first appendix summarises the checks that support the estimates in the main text; the appendices that follow contain the supporting material. Appendices B and C document the measurement of the supply variables and the treatment of the most recent data. Appendix D reports the counterfactuals of Section 5.5 as gaps from the observed paths; Appendices E, H and F display the imposed LNG paths and the demand shock switched off; Appendix G separates weather from autonomous saving in that shock; and Appendices I and J report two variants of the counterfactuals. Appendix K adds credible bands to the crisis counterfactuals of Section 5.6, and Appendix L a forward-looking scenario exercise for the Hormuz episode. Appendix M reports the full grid of impulse responses. Appendices N and O collect the daily and corroborating evidence behind the storage and LNG-supply priors, and Appendix P re-estimates the model on the IEA trade data.

Convergence. A second chain of 2 million draws from a dispersed starting point and a different seed reproduces every posterior median to the second decimal and the twelve-month precautionary share to within one percentage point. Effective sample sizes range from above 3,000 for β_{qp} and above 1,500 for ψ_3 to roughly 200 where the ridge runs, for γ_{qp} , ψ_1 and ψ_2 ; Geweke statistics are unremarkable throughout, and posterior statements about locations along the ridge should be read with that thinning in mind.

Seasonal treatment of consumption. Adjusting the consumption series with X-13 ARIMA-SEATS, as in the baseline, and leaving it unadjusted with monthly dummies in the deterministic block moves the posterior only modestly: γ_{qp} from 0.116 to 0.130, α_{qp} from -0.098 to -0.125 , and the twelve-month precautionary share from 47% to 42%, with β_{qp} and the aggregation weights unchanged.

Prior sensitivity. The first check is the one the measured priors exist to answer, and we report it in both directions. Re-estimating with agnostic scales of 0.5 on the storage couplings, the specification this paper would have reported absent the daily evidence of Appendix N, moves γ_{qp} from 0.116 to 0.197, ψ_1 from -0.20 to -0.93 , and the twelve-month precautionary share from 47% to 41%, while β_{qp} , α_{qp} and the aggregation weights stay put. Loosening the prior scale on α_{qp} fivefold moves its median to -0.057 , toward smaller elasticities rather than larger ones, and the precautionary share to 52%: the inelastic demand that concentrates price variance in storage demand is the likelihood’s choice, not the prior’s. Re-weighting the posterior draw by draw over a class of storage-block priors around the measured ones, with scales from one to four times the measured standard errors and shifted centres for ψ_2 , keeps the twelve-month share within [40%, 50%]. Pushing the measured priors to their dogmatic limit, scales of 0.01 that effectively impose $\psi_1 = \psi_3 = 0$, continues the same pattern rather than breaking it: γ_{qp} falls to 0.084, the precautionary share rises to 52%, and the probability that LNG is the more elastic channel reaches 0.87. The baseline sits between the agnostic and dogmatic ends at the point the daily evidence warrants, and its headline figures are the conservative ones. Prior variation, in short, does not move the result; measurement does. Replacing the supply data with the IEA aggregates places the share at 26%, and we regard the pair [26%, 47%] as the honest measurement band, with Appendix P discussing why the two databases disagree. The balance weight w_S remains likelihood-dominated in

the strongest sense: re-estimating with its prior deliberately mis-centred at 1.2 with scale 0.15, the posterior returns to 1.71, more than three prior standard deviations away, with γ_{qp} unmoved and α_{qp} shifting only modestly within its baseline interval.

Composition of the precautionary shock. Because this shock is identified through the storage equation and carries much of the price variance, one may ask whether it absorbs supply shocks rather than demand for stocks. The recovered series behaves as a structural innovation should: it is serially uncorrelated ($\rho_1 = -0.01$), contemporaneously orthogonal to both supply shocks (correlations -0.07 and $+0.02$), and unpredictable from four lags of either supply shock ($R^2 = 0.006$, $p = 0.999$); its correlations with future realisations of both supply shocks are likewise negligible. Its largest realisations fall on the recognisable expectational episodes of the sample: the March 2022 invasion month, the March 2026 closure of the strait, the July 2022 run-up ahead of winter, the January 2020 demand collapse and the pre-winter storage scramble of September 2021. What the shock does absorb, by construction, is the expectational component of supply news: an event that shifts desired inventory holdings is recorded here as precautionary demand, in exactly the sense in which Kilian (2009) attributes the 1990–91 Gulf War price spike almost entirely to precautionary demand. The label records the channel of operation, not the ultimate trigger, and the statements in this paper should be read accordingly.

Storage-variable convention. The baseline measures storage as the log change of the stock. Re-estimating with inventories entering instead as the change in stocks scaled by lagged supply, the convention of Rubaszek et al. (2021) and Farag (2024), leaves the precautionary share of the twelve-month price variance at 44% against 47% in the baseline, and the balance weight w_S lands on its recalibrated theoretical value near one. The importance of storage is not an artefact of how the storage variable is transformed.

Alternative data source. Appendix P re-estimates the model with the two supply quantities replaced by the IEA *Gas Trade Flows* aggregates. The LNG supply elasticity falls to 0.035 and the pipeline elasticity rises to 0.63, so that the ordering of the two channels reverses. We interpret this as a measurement result rather than as a fragility of the model: the IEA LNG series records scheduled terminal send-out, which by construction cannot respond to a price within the month, whereas the JODI series records cargo arrivals, which can. The elasticity attributed to a supply channel is therefore a property of the measurement concept as much as of the market.

B Measurement details of the supply variables

This appendix collects the measurement detail behind the two supply aggregates that Section 4.2 summarises: which point of the LNG chain the arrivals variable measures, the buffers that sit between arrival and delivery, and the reasons for keeping pipeline imports and domestic production in a single aggregate.

It matters for the interpretation of the estimates which point of the LNG chain the variable q_{LNG} measures. Our data record LNG as it arrives aboard ocean tankers, on a regasified-equivalent basis: q_{LNG} measures cargo receipts, not the volume vaporised and delivered into

the transmission grid during the same month. The two differ by a buffer. LNG discharged from a vessel is held as a liquid in cryogenic tanks at the terminal, not regasified on arrival, and is vaporised into the grid on a schedule chosen by the holders of regasification capacity, subject to a technical minimum send-out and to the terminal’s maximum hourly capacity. Terminal tankage typically corresponds to between a few days and two or three weeks of send-out, short relative to a year but not relative to a month. A second buffer sits one step earlier: a laden vessel may wait offshore rather than discharge, using the ship itself as floating storage, as happened on a visible scale in late 2022 when European terminals were operating at saturation.

Neither buffer is observed by the model. Our inventory variable s is JODI’s closing stock level, which covers recoverable gas held in storage facilities and excludes cushion gas, line pack and LNG held at import terminals. The system therefore contains cargo arrivals and underground gas storage but not the terminal and floating inventory that sits between them, and whatever timing discretion is exercised in that gap is absorbed into the estimated dynamics of q_{LNG} . We return to this point when interpreting the LNG-supply elasticity in Section 5.1 and when comparing data sources in Appendix P, where the alternative source measures the other side of the same buffer.

We deliberately keep q_{pip} as a single aggregate rather than introducing domestic production as a third VAR variable. What separates the two supply channels in this model is institutional: pipeline imports and domestic production are both tied to long-term contracts, fixed infrastructure and regulated extraction paths, so within the month their volumes follow commitments rather than the spot price, whereas LNG cargoes are traded globally and can in principle be redirected. On that dimension domestic production belongs with pipeline imports, not with LNG. Adding a third supply variable would consume identification room without delivering an economically distinct shock, and the resulting series would be dominated by the Groningen-related structural decline rather than by short-run market dynamics.

C Data treatment at the bleeding edge of the sample

The estimation sample ends in May 2026 and is built from the JODI Gas vintage released on 19 July 2026. The Hormuz crisis began on 28 February 2026, so March, April and May 2026 are the first three full months of the episode and all three are inside the sample.

The maturity of the data differs across these three months, and the pattern is the one JODI’s revision cycle implies. March and April 2026 are fully mature: every one of the seven flows we use (INDPROD, IMPPIP, EXPPIP, IMPLNG, EXPLNG, CLOSTLV, STOCKCH) carries assessment code 1 (*Final*) for all 22 member states in the panel. Only the final month, May 2026, still contains preliminary entries, and they are confined to two groups.

First, four small countries (EE, IE, LU, SI) report the monthly stock change (STOCKCH) with code 3 (*Imputed*) in May 2026. Rather than use JODI’s own imputation, we reconstruct their stock change from the country-level first-difference of the closing stock level (CLOSTLV),

$$\widehat{\text{STOCKCH}}_{c,t} = \text{CLOSTLV}_{c,t} - \text{CLOSTLV}_{c,t-1},$$

an accounting identity that uses only data reported with code ≤ 2 by the national authorities of those four countries. They jointly represent less than 3% of EU-27 gas consumption, so the quantitative effect on the aggregate `STOCKCH` series is negligible.

Second, four larger countries submit some flows with code 2 (*Preliminary*) in May 2026: industrial production for BE and NL, pipeline and LNG exports for DE, and all four trade flows for FI. We include these because JODI’s standard practice reclassifies preliminary entries to Final in subsequent monthly releases, and historical revisions between Preliminary and Final vintages have been small over our sample.

The consumption series raises a separate timing issue. Eurostat’s `nrg_cb_gasm` is published with a lag of roughly two months and is revised for several vintages thereafter; the May 2026 observation used here is a first release. Because consumption enters in log growth rates and the counterfactual and decomposition exercises are anchored well before the end of the sample, a revision to the final month would not alter any reported result. The series also determines the sample start: Eurostat publishes it from January 2014, which is why the baseline is estimated from February 2014 while a specification without consumption could start in February 2010.

The deflator raises no comparable issue in this vintage. The euro-area HICP all-items index is published through June 2026 and every observation used here is a published figure. The World Bank Pink Sheet gas quotation, a monthly average of daily prices, is likewise available through June 2026. Since the price enters the system in log growth rates and monthly HICP inflation has a standard deviation of 0.53 percentage points against 14.2 for the gas price itself, the deflation step accounts for well under one percent of the variance of the price series.

The empirical results presented in the main text are robust to dropping the last observation; we report the full sample because the Hormuz episode is precisely the event we want the estimated dynamics to incorporate.

D The counterfactual gaps

Figure 4 reports the three counterfactuals as log-level paths against the observed series. Figure 6 reports the same exercises as gaps, the counterfactual path minus the observed one, in log points and with the same 68% and 90% bands. The gap representation makes the magnitudes and their uncertainty directly readable, and the common scale within each row makes the three exercises comparable: closing the LNG channel and opening it fully move the price by roughly equal and opposite amounts, while removing the demand reduction moves it by about three times as much.

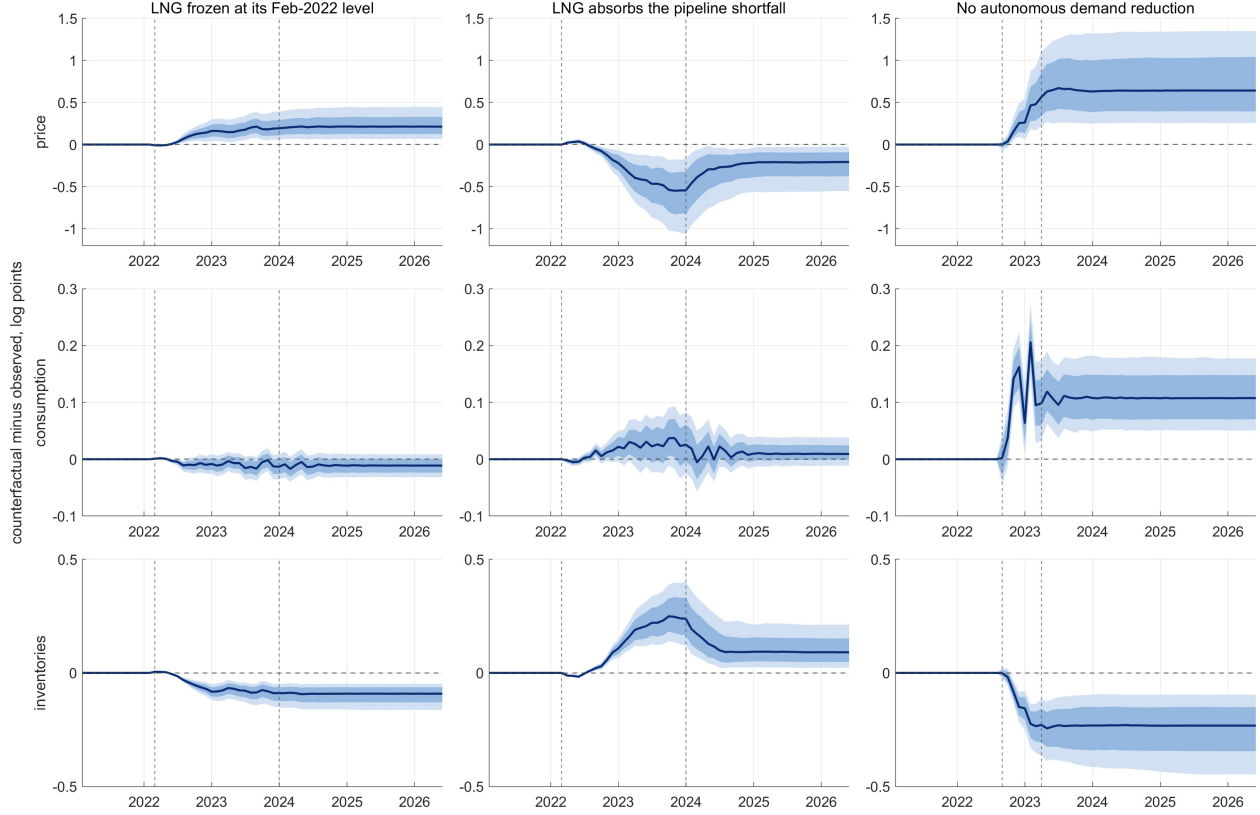


Figure 6: The three counterfactuals as gaps. Columns and constraint windows as in Figure 4; each panel shows the posterior median with 68% and 90% credible bands of the difference between the counterfactual and the observed log level, in log points. Rows share a common scale across columns.

E Imposed LNG path under Counterfactual 1

Figure 7 reports the imposed path of net LNG imports under Counterfactual 1 against the observed series. The left panel shows the level, indexed to the February 2022 anchor month: the counterfactual remains flat at the freeze value while the observed series continues to grow. The right panel shows monthly log-growth, which the constraint sets to zero throughout the window. After December 2023 the constraint is released and the structural innovations to the LNG-supply equation return to their historical values; the path is not forced back onto the observed series, so the post-window gap reflects the accumulated dynamic response of the system rather than a mechanical rejoin.

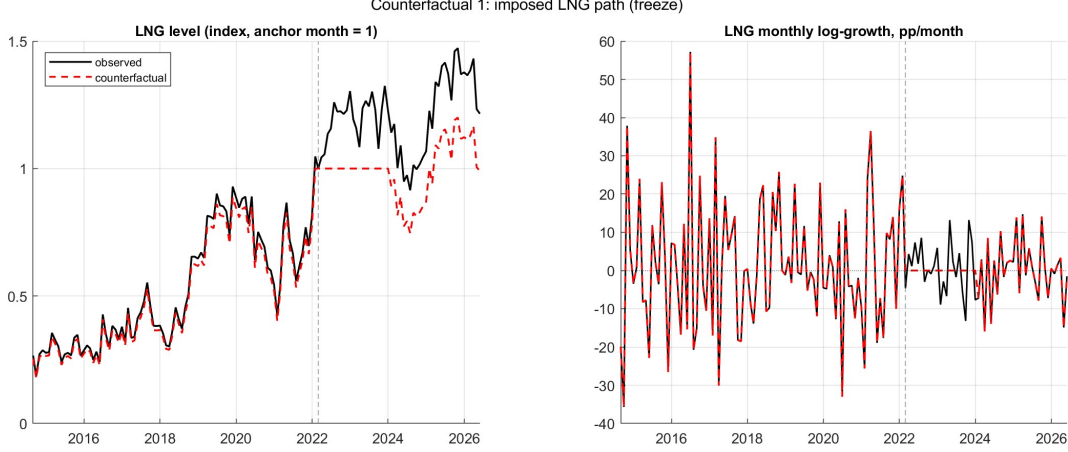


Figure 7: Counterfactual 1: imposed path of net LNG imports. *Left:* level indexed to the anchor month, observed (solid black) versus counterfactual (dashed red). *Right:* monthly log-growth in percentage points, observed versus counterfactual. The vertical dashed line marks February 2022, the start of the freeze window.

F The structural demand shock u^3 under Counterfactual 3

Figure 8 reports the flow-demand shock u^3 as identified by the six-variable model, together with its counterfactual replacement. In this specification u^3 is the innovation of a demand equation written for observed consumption, which is what makes Counterfactual 3 an exercise about the demand for gas rather than about a constructed residual.

The left panel shows the posterior median of u^3 across the retained draws with its 5–95% credible band; the right panel shows the cumulative sum, which makes the persistence of the autonomous demand reduction visible. During the policy window the median takes systematically negative values, the empirical signature of the joint policy, temperature and behavioural-conservation response discussed in Section 5.5.3. That this shock correlates at 0.75 with heating degree days is a reminder that the model identifies an autonomous movement in consumption but cannot, on its own, apportion it between policy and weather.

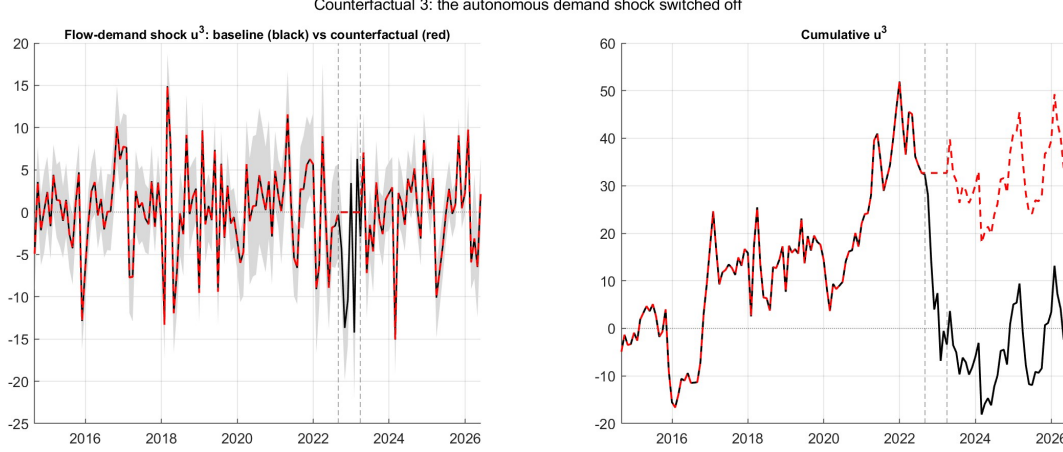


Figure 8: Counterfactual 3: identified flow-demand shock u^3 . *Left*: posterior median over the sample (solid black) with its 5–95% credible band, and the counterfactual replacement (dashed red, zero within the policy window). *Right*: cumulative sum of the same two series. The vertical dashed lines mark 1 August 2022 and 31 March 2023, the boundaries of the formal policy window under Council Regulation 2022/1369.

G Separating weather from autonomous saving in the flow-demand shock

The flow-demand innovation bundles weather-driven consumption with the autonomous saving of gas. Because weather is exogenous, the two can be separated by a projection. Within each posterior draw, the flow-demand innovations u_t^3 recovered by that draw are regressed, by ordinary least squares over the full estimation sample, on a constant and the monthly heating-degree-day anomaly, the deviation of population-weighted European heating degree days from their calendar-month average. An unusually cold month raises the demand for gas mechanically, through heating, and the anomaly cannot itself respond to the gas market, so the part of the shock that covaries with the anomaly is attributable to weather. The weather component of each month's shock is therefore the anomaly term of the fitted value, and the autonomous component is the rest, the least-squares residual together with the constant; the two add up to the shock by construction. Estimating the loading over the full sample rather than the window alone sharpens it, at the cost of assuming that the response of demand to a temperature surprise is stable over time. Each component is then propagated through the same draw's price responses, and its cumulative contribution to the log price of gas over the window of Council Regulation 2022/1369, August 2022 to March 2023, is recorded; because every step runs within the draw, the split inherits the posterior uncertainty of the model.

Over that window, flow-demand shocks contributed -0.71 log-points to the price, with a 90% credible interval of $[-1.25, -0.26]$. The weather component accounts for -0.22 of that, with an interval of $[-0.35, -0.10]$, and the autonomous component for -0.48 , with an interval of $[-0.93, -0.11]$: roughly one third and two thirds of the total. The autonomous component is the object that the counterfactual of Section 5.5.3 speaks to, and within it the Regulation cannot be separated from voluntary conservation. The split nevertheless bounds the role of

luck: even had the winter been normal, about two thirds of the demand-side relief would have remained.

H Imposed LNG path under Counterfactual 2

Figure 9 reports the imposed path under Counterfactual 2, anchored at the February 2022 LNG volume and rising one-for-one with the contemporaneous pipeline-and-domestic short-fall,

$$\text{LNG}_t^{cf} = \text{LNG}_{\text{Feb-22}} + [q_{\text{Feb-22}}^{pip} - q_t^{pip}].$$

The imposed level peaks at 1.99 times its February 2022 value during the worst of the crisis. Those volumes exceed the regasification, downstream-pipeline and global cargo availability that were operationally feasible in 2022–2023, which is precisely the physical bottleneck the scenario abstracts from. A single rejoin month in January 2024 returns the LNG level to the observed series; the sharp movement visible in the growth panel at that date is a mechanical consequence of that rejoin and not a feature of the counterfactual path itself.

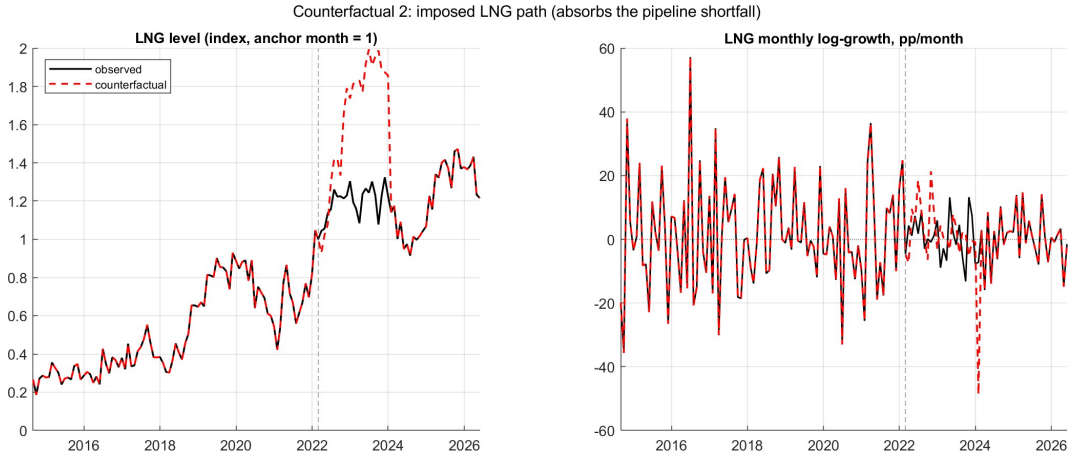


Figure 9: Counterfactual 2: imposed path of net LNG imports. *Left*: level indexed to the anchor month, observed (solid black) versus counterfactual (dashed red). *Right*: monthly log-growth in percentage points. The vertical dashed line marks February 2022, the start of the absorption window.

I Counterfactual 1b: LNG growing at its pre-crisis trend

Counterfactual 1 freezes LNG imports, a deliberately extreme bound. A milder variant constrains their growth not to zero but to the pre-crisis trend $\bar{g}^{LNG}$, computed as the mean monthly log-growth of net LNG imports over January 2017 to December 2021, which is 1.28 percentage points per month or roughly 16% annualised. The constraint is imposed over the same February 2022 to December 2023 window by the same procedure. The exercise separates the contribution of the crisis-induced policy push, floating storage and regasification

units, the EU–US task force, accelerated terminal commissioning, from the secular expansion already under way.

The real price ends the sample about 13% below the observed path, with consumption 0.7% higher and inventories 5.9% higher. The sign deserves explanation. Applying the pre-crisis trend over twenty-two months delivers cumulative LNG growth of about 28 percentage points, whereas the realised cumulative growth over the same window was smaller. The trend counterfactual therefore puts more LNG into the system than was actually achieved, and prices fall accordingly. Two readings follow. First, the realised expansion during 2022–2023 fell short of what continuing the pre-crisis trajectory would have delivered, so the crisis-induced interventions were necessary at the margin rather than redundant. Second, the magnitudes are modest either way: the frozen-LNG bound of Counterfactual 1 leaves the price roughly a quarter higher, so the entire span between trend continuation and complete unavailability is around 40% of the terminal price. The stabilising contribution of LNG came primarily from receiving infrastructure accumulated over the preceding decade.

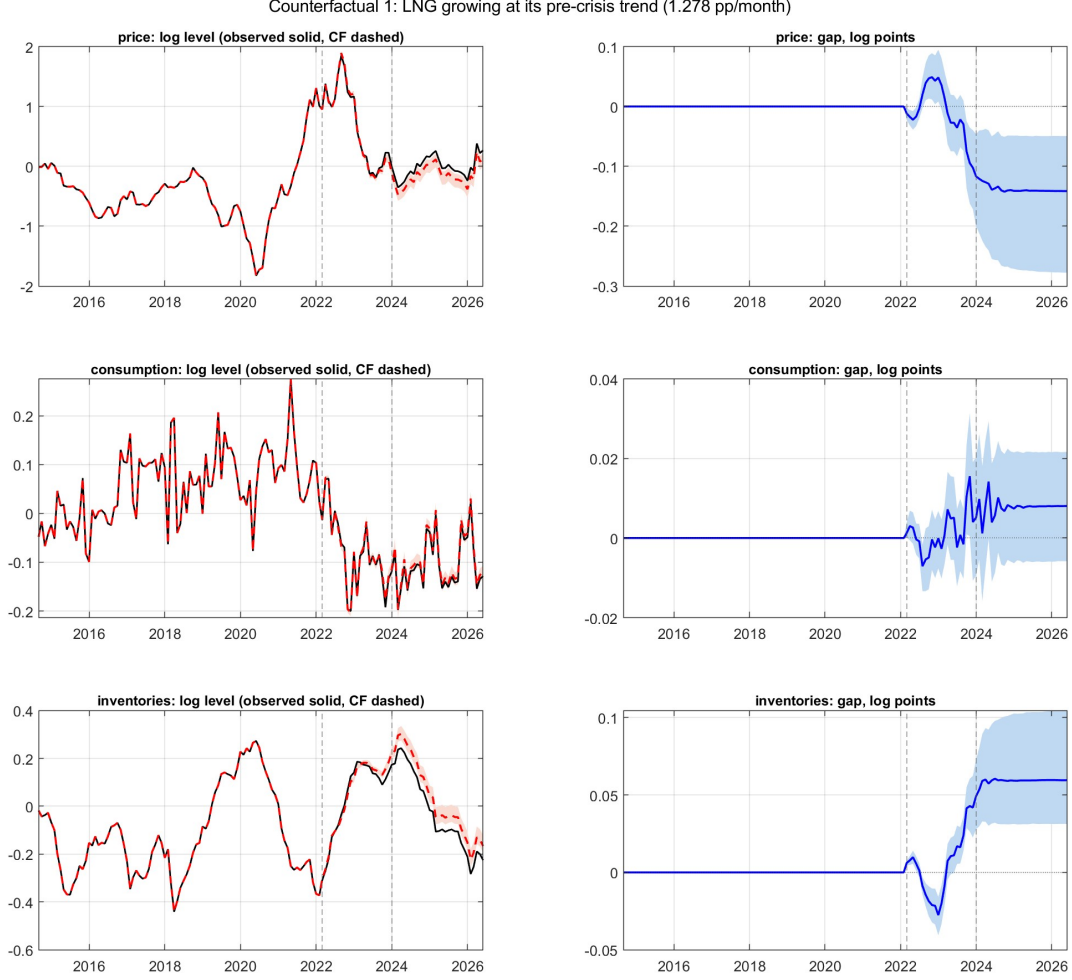


Figure 10: Counterfactual 1b: response of the real price, observed consumption and inventories under $g_t^{LNG} = \bar{g}_{\text{pre-crisis}}^{LNG}$. *Left column*: log-level, observed (solid black) and counterfactual median (dashed red) with the 5–95% credible band. *Right column*: the posterior gap in log-points. Vertical dashed lines mark the constraint window.

J Counterfactual 3 over an extended window

The baseline of Counterfactual 3 switches off the flow-demand shock over the formal window of Council Regulation 2022/1369, August 2022 to March 2023. The Regulation was extended by Council Regulation 2023/706 through March 2024, and the demand reduction it targeted persisted well beyond the original window. Figure 11 repeats the exercise over August 2022 to December 2023. The qualitative pattern is unchanged and the magnitudes are larger, as the longer window removes a longer sequence of autonomous demand reductions. We report the shorter window in the main text because it corresponds to the period in which the policy target was formally binding, and because attributing the whole of the extended period to an autonomous demand response would strain the bundling caveat discussed in Section 5.5.3.

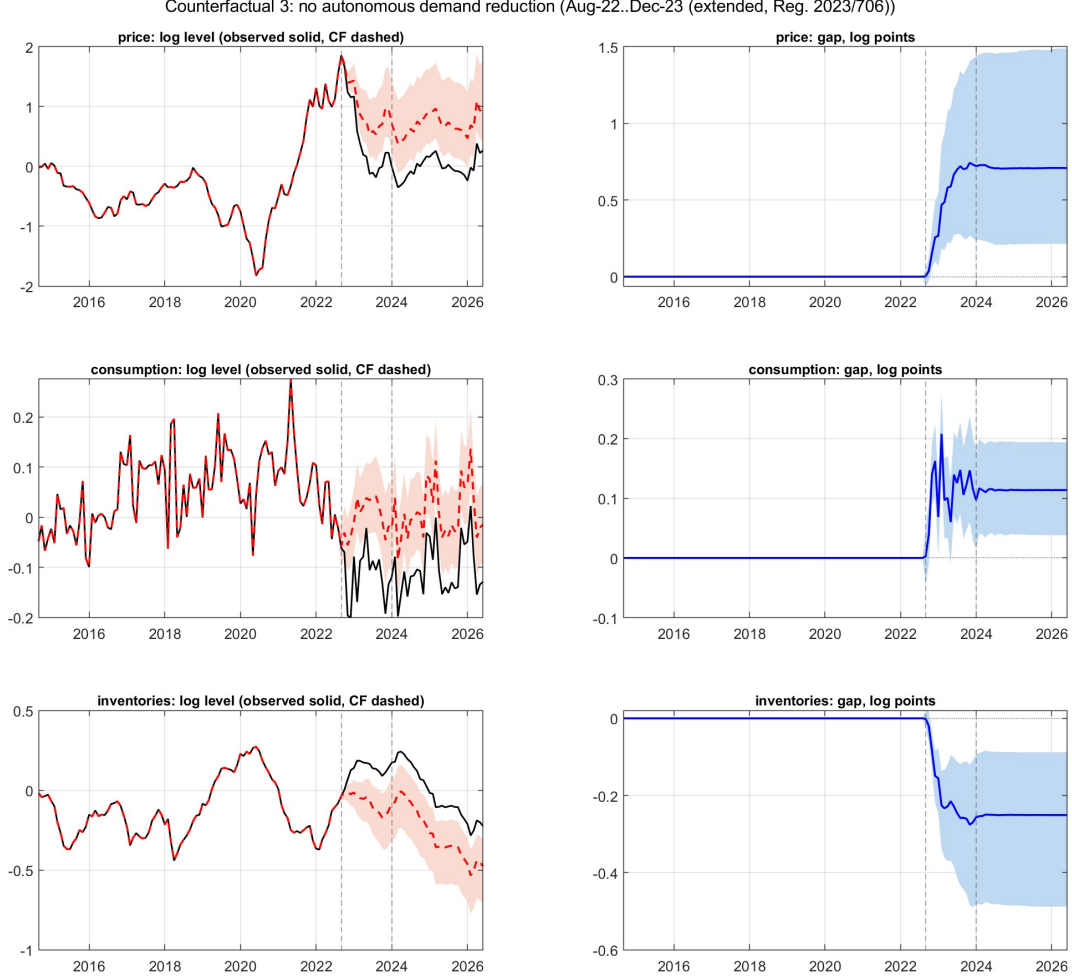


Figure 11: Counterfactual 3 over the extended window: response under $u_t^3 = 0$ from August 2022 to December 2023. Layout as in Figure 10. The two vertical dashed lines mark the boundaries of the extended constraint window.

K Credible bands for the crisis counterfactuals

Figure 5 reports posterior medians. Figure 12 repeats the exercise one channel per panel, with the 68% and 90% credible bands of each counterfactual path. Two features are worth reading off. The width of a band grows with the size of the removed contribution, so the path without precautionary shocks is the most uncertain in both crises. And for the short 2026 window it is also the only counterfactual that separates decisively from the observed path: three months of data identify the split between channels less sharply than the point estimates alone suggest.

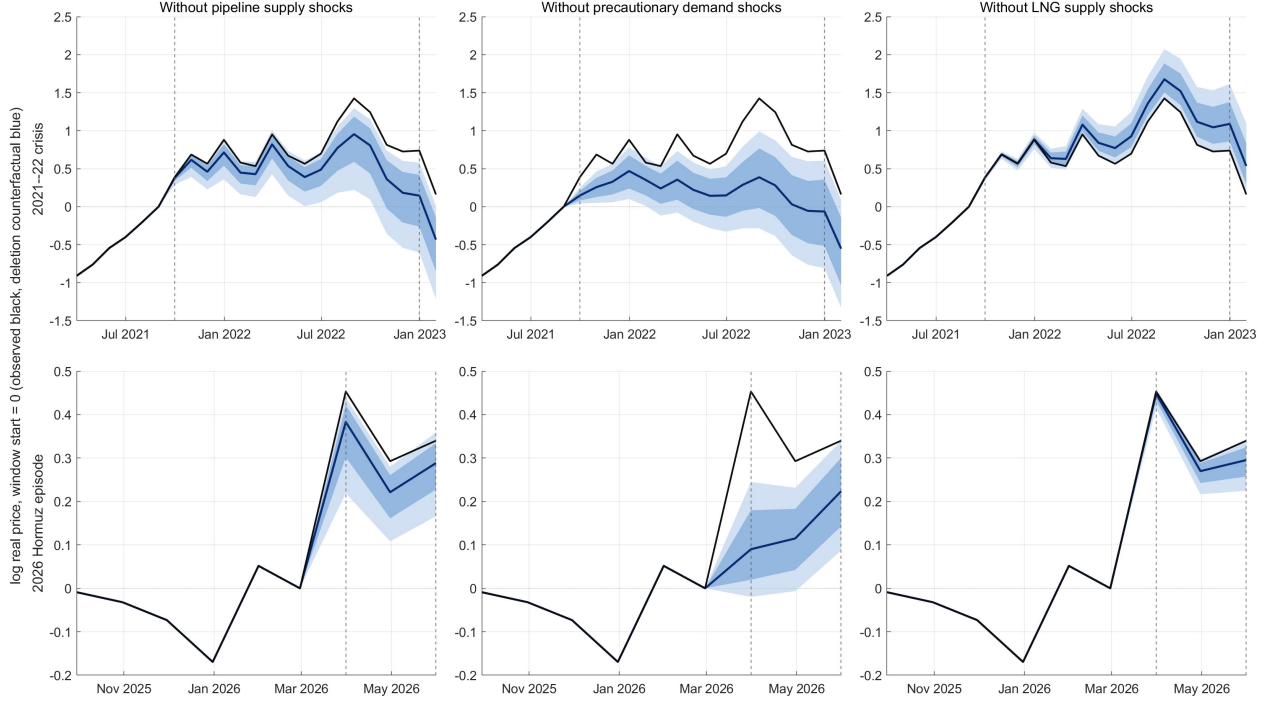


Figure 12: The crisis counterfactuals with credible bands. Rows are the two crises, columns the removed channel. Each panel shows the observed log real price, normalised to zero in the month before the window opens (black), and the posterior median of the counterfactual without that channel’s shocks (blue) with 68% and 90% credible bands. Vertical dashed lines mark the crisis window; panels share scales within each row.

L A forward-looking two-channel scenario fan for the Hormuz episode

The retrospective analysis of Section 5.6 uses only recovered innovations and imposes nothing. This appendix reports a complementary forward-looking exercise, which we present as an illustration of the transmission mechanism rather than as inference: its scenario paths are calibrated by us, and any forward projection of an ongoing geopolitical episode dates quickly.

A Hormuz event in the language of this model is a joint path of two structural shocks, because that is what the realised episode looks like. In March 2026, when the price rose 45%, European LNG arrivals rose by 3.4% and the precautionary innovation was large and positive; in April, when arrivals fell 14.9%, the truce unwound the fear premium and the price fell. Scenarios therefore impose a path for the LNG-supply innovation u^5 and a path for the precautionary innovation u^4 , each scaled per draw by the magnitude of that draw’s realised March 2026 shock of the same type. The projection runs from June 2026 to May 2027; June and July are common across scenarios and calibrated to the realised narrative, and the three scenarios, de-escalation, frozen conflict, winter escalation, diverge from August.

Because the model is linear, the total price gap decomposes exactly into the two channels,

and that decomposition is the point of the exercise. Under the winter-escalation scenario the peak gap of 1.22 log-points, reached in January 2027, decomposes into 0.06 log-points from the volume channel and 1.13 from the expectational channel, with 0.03 of interaction: the expectational channel accounts for 93% of the effect and the volume channel for about 5%. Running the volume channel alone, the exercise an aggregated specification would naturally suggest, produces a peak gap of 0.07 log-points, a fan so flat as to imply that a sustained closure of the strait would barely register in European prices. That conclusion would be an artefact of omitting the channel through which the realised episode actually operated.

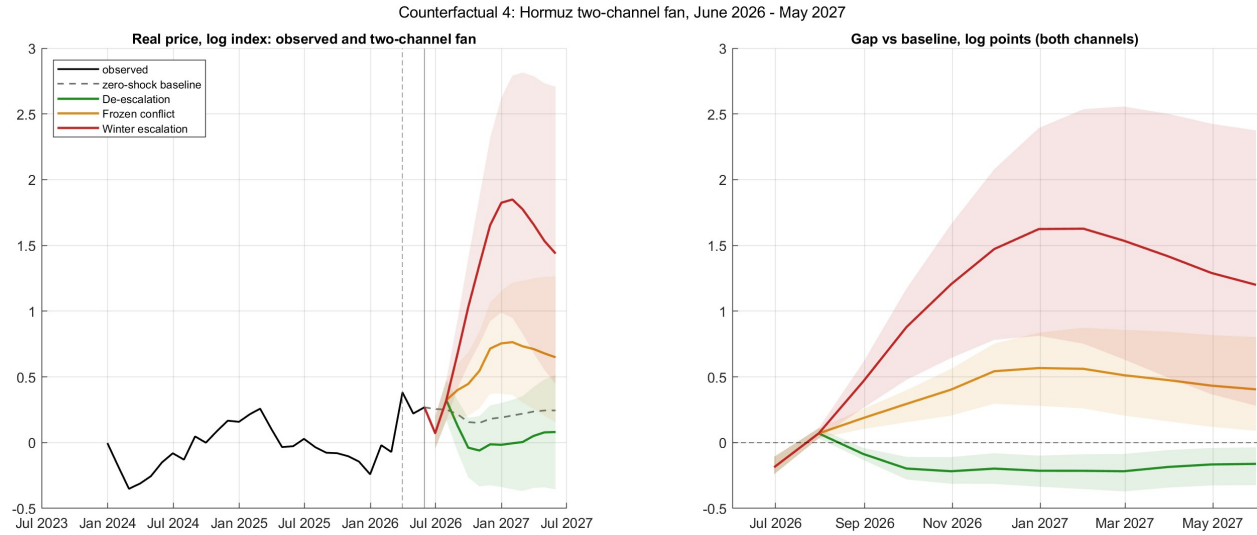


Figure 13: Two-channel scenario fan. *Left:* real price in log index, observed over the last thirty months of the sample (solid black), the zero-shock baseline projection (grey dashed) and the three scenario medians with their 5–95% credible bands. *Right:* the gap against the zero-shock baseline in log-points. The vertical lines mark the end of the estimation sample and March 2026.

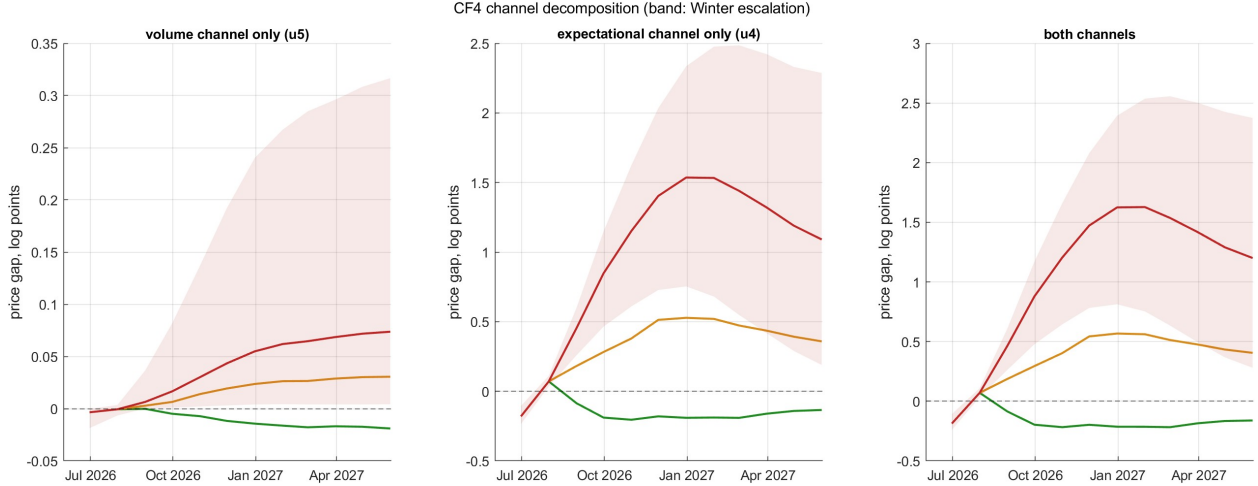


Figure 14: Channel decomposition of the fan. *Left*: the volume channel alone, imposing only the LNG-supply innovation. *Centre*: the expectational channel alone, imposing only the precautionary innovation. *Right*: both channels. Lines are posterior medians for each scenario; the shaded band is the 5–95% interval for the winter-escalation scenario. The near-flatness of the left panel against the centre panel is the substantive content of the figure.

M The full grid of impulse responses

Figure 15 reports the cumulative responses of all six endogenous variables to all six structural shocks, completing the price responses shown in Section 5.1. As in the main text, the two supply shocks are displayed as adverse.

The dynamic contrast between the two supply channels deserves separate comment. The posterior probability that the cumulative response of LNG arrivals to a flow-demand shock exceeds that of pipeline supply ranges between 0.50 and 0.73 across horizons: with the demand shock identified from observed consumption, the dynamic version of the flexible-versus-rigid characterisation leans the same way as the contemporaneous one but is not established either.

N Daily evidence for the storage couplings

This appendix documents the daily evidence behind the measured priors on ψ_1 and ψ_3 of Section 4.4. Four daily series are used. European storage stocks and net injections are published by Gas Infrastructure Europe from January 2014 onward without gaps, and the send-out and inventories of the regasification terminals come from the same platform’s LNG transparency service from 2012. External pipeline arrivals are the sum of physical entry flows at the external border points reported on the ENTSOG transparency platform, covering the Norwegian, Russian, North African and Azerbaijani corridors together with the two British landing points; the platform’s records begin in 2016, which sets the start of the estimation

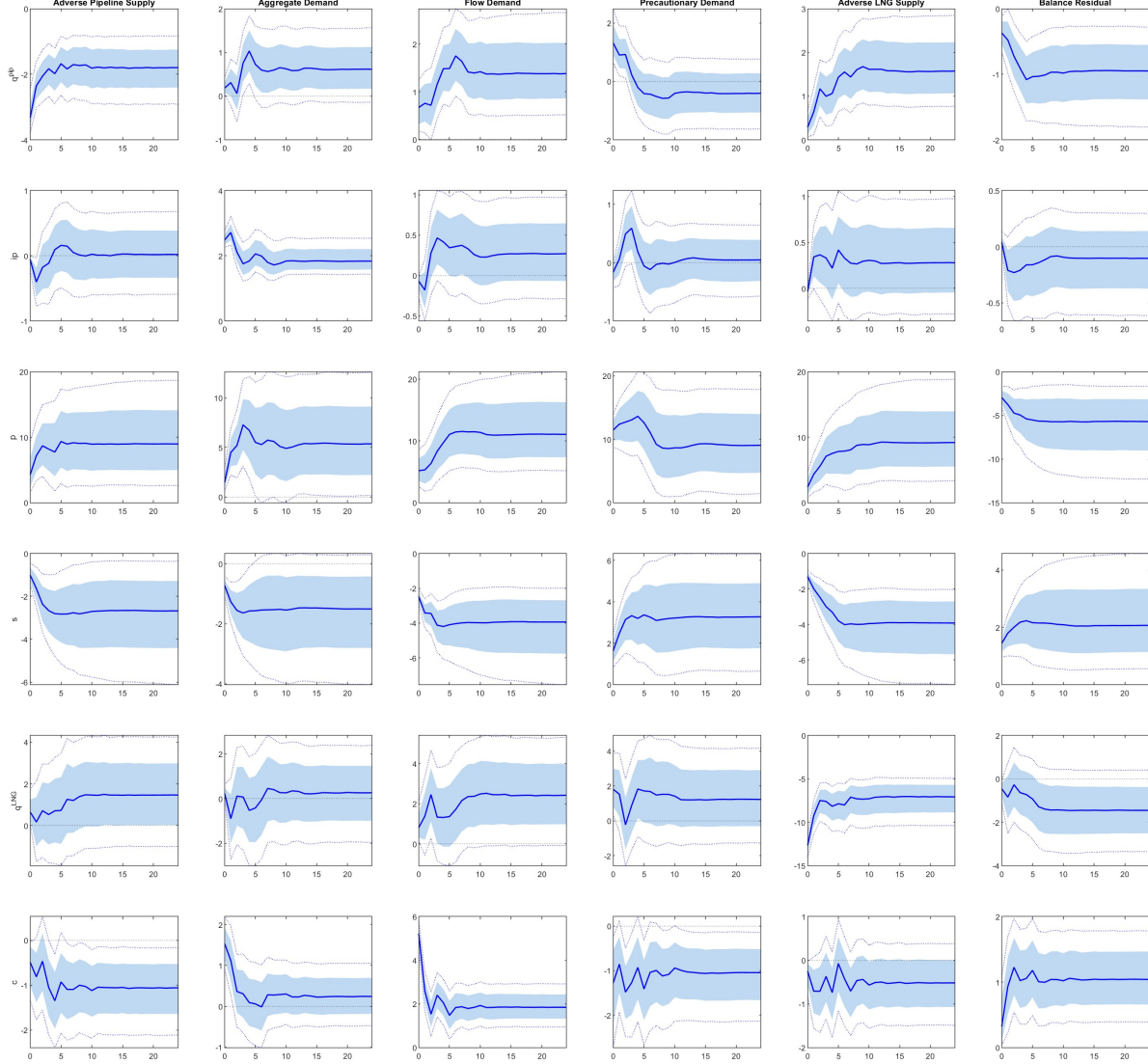


Figure 15: Cumulative impulse responses of the six endogenous variables (rows) to the six structural shocks (columns). The solid blue line is the posterior median, the shaded area is the 68% credible band and the dotted lines the 90% band. The horizon is 24 months. All responses are cumulated.

sample.¹¹ The controls are the TTF day-ahead settlement price, carried forward over non-trading days, and population-weighted European heating degree days.

The case for measuring these priors starts from what the parameters are. A monthly observation aggregates thirty daily decisions of the same storage operators, and those decisions are observable, so the within-month comovement that ψ_1 and ψ_3 summarise can be computed directly instead of being treated as unknown. Left agnostic, the couplings are resolved

¹¹Duplicate reporting by co-owning transmission operators at the German landing points was identified and removed before aggregation, and the aggregate was validated against known route-level volumes. Because one British entry point mixes Norwegian gas with United Kingdom continental-shelf production, the daily specification below was re-estimated excluding both British points, with a same-day coefficient of 0.17 against 0.13 in the baseline aggregate.

jointly with the supply elasticities along posterior ridges: under scales of 0.5, the posterior correlation is -0.75 between ψ_1 and γ_{qp} and -0.64 between ψ_3 and β_{qp} , so the monthly likelihood pins down combinations of a supply elasticity and a storage coupling rather than each parameter separately.

The monthly specification is the direct counterpart of the couplings in the storage equation. The growth of the end-of-month stock is regressed on the growth of monthly pipeline arrivals and of monthly LNG send-out, controlling for the growth of the average price, heating degree days and month-of-year effects, with Newey–West standard errors. Over February 2016 to June 2026 the estimates are as follows.

	Coefficient	Std. error	90% interval
Pipeline arrivals (counterpart of ψ_1)	0.015	0.065	$[-0.092, 0.122]$
LNG send-out (counterpart of ψ_3)	0.009	0.035	$[-0.048, 0.066]$
Real price	-0.015	0.029	
Heating degree days	-0.0014	0.0004	

Table 7: Monthly evidence regression for the storage couplings. Dependent variable: growth of the end-of-month EU storage stock. Month-of-year effects included; Newey–West standard errors with six lags; $n = 125$, $R^2 = 0.90$, February 2016 to June 2026.

Both coefficients are zero within tight bounds, and the conclusion survives every variant we have examined. Measuring the pipeline coupling alone, without the LNG regressor, gives 0.012 with a standard error of 0.065. Measuring LNG as reconstructed ship arrivals, the sum of send-out and the change in terminal inventories, gives -0.007 with a standard error of 0.007. Extending the LNG regression back to 2014 on the longer storage sample, without the pipeline control, gives 0.004 with a standard error of 0.035. Figure 16 displays the two relationships with the controls partialled out and the crisis months marked, showing that the absence of comovement is a feature of the full sample rather than an artefact of the crisis.

At daily frequency the same series do show a link. Regressing daily net injection on same-day arrivals through both channels, with the price, heating demand, day-of-week, month-of-year and year effects and one lag of the dependent variable, gives same-day coefficients of 0.16 for pipeline arrivals and 0.15 for LNG send-out, each with a standard error of 0.03; for pipeline arrivals a sustained increase cumulates to roughly one half of the extra flow once the injection process has adjusted. The contrast between an operational same-day response and a monthly elasticity of zero is informative rather than contradictory. Injections route part of the gas landing on a given day, but over a month the stock position is set by the seasonal cycle, the price and heating demand, and does not track the month’s arrival volumes. The couplings the model requires are the monthly ones.

These regressions are descriptive, and the direction of the remaining simultaneity works against the conclusion rather than for it: a storage-demand shock that attracts additional arrivals raises the measured comovement, so the coefficients of Table 7 are, if anything, upper bounds on the size of the structural couplings. The priors of Section 4.4 accordingly keep the centres at zero and set the scales to twice the Newey–West standard errors, 0.13 for ψ_1

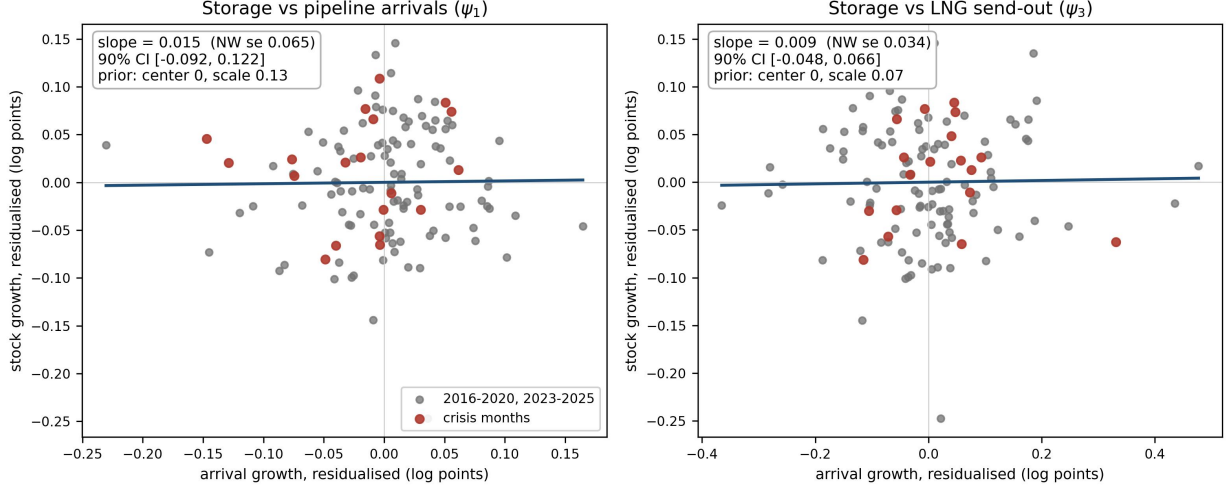


Figure 16: Daily-data evidence behind the measured priors on the storage couplings. Each panel plots monthly EU stock growth against the growth of one arrival aggregate, both residualised on the other aggregate, the real price, heating demand and month-of-year effects, over 2016–2026. By construction the slope of the fitted line equals the corresponding coefficient of the joint regression, 0.015 (Newey–West standard error 0.065) for pipeline arrivals and 0.009 (0.035) for LNG send-out. Crisis months (September 2021 to December 2022 and March to May 2026) are marked in red.

and 0.07 for ψ_3 , a choice that is agnostic about sign, honest about specification uncertainty, and wide enough for the likelihood to overrule the evidence where it insists. The posterior of ψ_1 indeed ranges beyond the prior scale where the data insist, and Appendix A reports the model re-estimated under the agnostic scales.

O Corroborating evidence on the LNG-supply elasticity

The LNG-supply elasticity is set-identified within the model, so it is worth asking what can be said about its size without the model’s identifying assumptions. This appendix reports three calculations, all computed on reduced-form innovations of the real price and of LNG arrivals, that is, on residuals of unrestricted autoregressions that use no sign restrictions and no priors.

The first is the moment itself. A contemporaneous supply elasticity links arrival innovations to price innovations, so a large elasticity requires the two series to covary within the month. Over the estimation sample their correlation is 0.001: the raw comovement from which any large contemporaneous response would have to be identified is essentially absent.

The second is a regime-based estimator in the spirit of Rigobon (2003). The variances of the innovations shift sharply across market regimes (a pre-crisis regime through mid-2021, the crisis regime through the end of 2023, and a post-crisis regime thereafter), and if the supply slope is stable across regimes, those shifts trace it out: each regime contributes one moment

condition linking the slope to that regime’s second moments. Concentrating the demand-side slope on a grid and solving the per-regime conditions in closed form yields a supply slope of 0.07. A bootstrap that resamples innovations within each regime gives a 95% interval of $[-0.08, 0.20]$, and the bootstrap probability that the slope exceeds 0.5 is numerically zero.

The third asks what the data would have to look like for a large elasticity to be admissible. Holding the pre-crisis moments fixed, a slope of one would require the crisis-period covariance between the price and arrival innovations to equal 659; the measured covariance is 16.1 with a standard error of 36.0, eighteen standard errors short. Values consistent with an order-of-magnitude separation between the supply channels require correspondingly more.

None of these calculations identifies the elasticity sharply, and none replaces the model. What they establish is that the moments of the data bound the contemporaneous response of arrivals to the European price well below the values that a strong reading of LNG flexibility would need, which is consistent with the posterior reported in Section 5.1.

P Robustness: IEA Gas Trade Flows in place of JODI

This appendix re-estimates the six-variable model with the two supply quantities replaced by the IEA *Gas Trade Flows* aggregates: pipeline-and-domestic supply by the IEA pipeline aggregate (Russia, Norway, Algeria, Iran and Turkey, all routes summed) and LNG by the IEA LNG aggregate. Identifying restrictions, priors, consumption series, seasonal treatment and sample are unchanged, so any difference in the posterior is attributable to measurement alone.

The two sources measure different points of the supply chain

The difference between the sources is not one of accuracy but of position along the chain. JODI collects national reports of gas received by the reporting country, and its LNG import series is defined as LNG arriving aboard ocean tankers on a re-gasified-equivalent basis, so the JODI LNG series records cargo receipts. IEA Gas Trade Flows instead collects volumes that have physically crossed a national boundary at a named entry point, with LNG terminals appearing as entry points carrying a maximum hourly flow capacity against which reported monthly volumes are validated. A capacity expressed as a flow rate is a regasification capacity, so the IEA LNG series is closer to send-out into the national transmission grid. The two sit on either side of the terminal storage buffer described in Appendix B.

The data bear this reading out. In monthly log growth the JODI LNG series is the choppier of the two, with a standard deviation of 13.9 against 10.7 percentage points and a first-order autocorrelation of -0.17 ; the IEA series is smoother and positively autocorrelated at $+0.31$. Negative autocorrelation is the signature of lumpy discrete arrivals, in which a heavy month is followed by a light one; positive autocorrelation is the signature of a scheduled flow. The correlation between the two growth rates is only 0.50.

Two further conventions differ. IEA Gas Trade Flows reports on a gross rather than a net basis and counts transit volumes as both an import and an export of the transiting country,

whereas our JODI aggregates are net of exports; this inflates the variance of the IEA pipeline series, whose monthly log-growth standard deviation is 7.2 percentage points against 4.4 for JODI. Geographic scope also differs: the IEA pipeline aggregate counts external producer-country routes only and therefore excludes EU domestic production, which the JODI baseline includes. That exclusion explains the divergence of the cumulative log-levels in Figure 17: the Dutch Groningen field fell from roughly 50 bcm per year in 2013 to a near-zero phase-out by 2024, a secular decline that drags on the JODI growth rate and is absent from the IEA aggregate by construction.

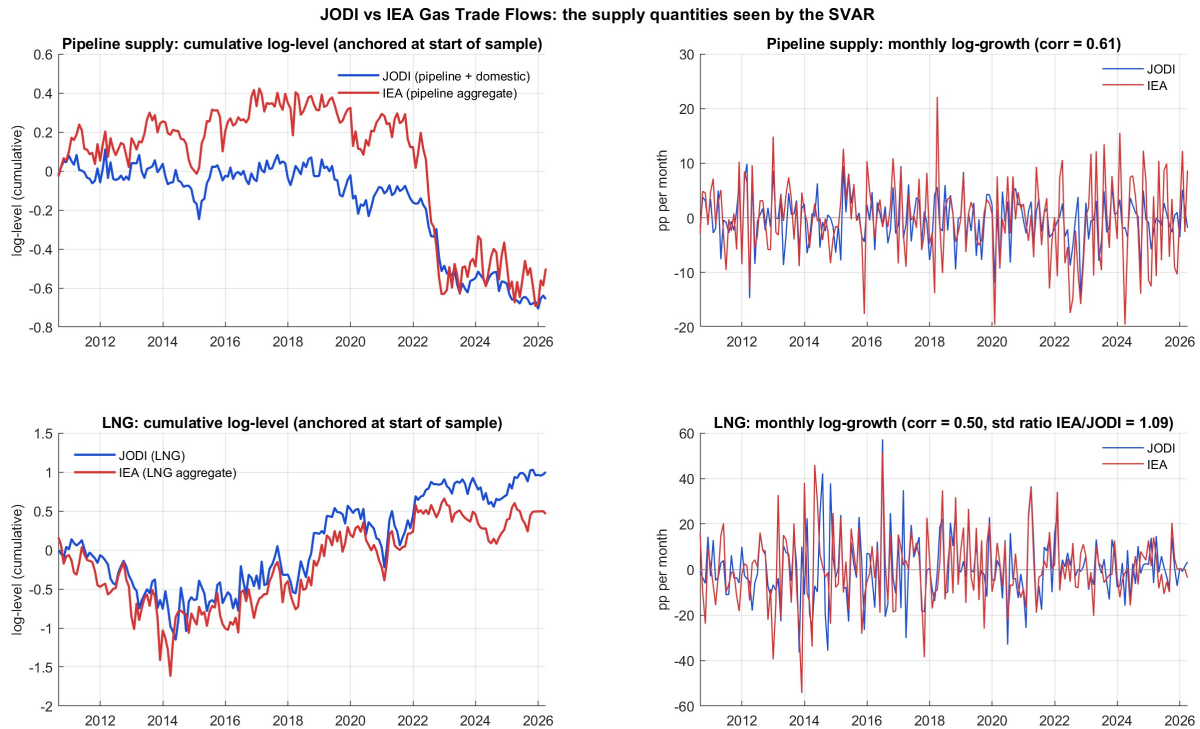


Figure 17: JODI baseline (blue) against IEA Gas Trade Flows (red) for the two supply aggregates. *Top row*: pipeline supply, in cumulative log-level and in monthly log-growth. *Bottom row*: LNG, in the same two representations. The IEA pipeline series is the more volatile in growth (std 7.2 against 4.4 pp); for LNG the ranking reverses (13.9 against 10.7 pp), and the autocorrelations have opposite signs, the pattern expected of lumpy cargo arrivals against scheduled terminal send-out.

Estimation

The IEA robustness exercise required more care in estimation than the baseline, and we report the difficulty because it bears on how the results should be read. The posterior mode under the IEA series sits on the sign-restriction boundary $\beta_{qp} \rightarrow 0$, at which the Hessian is not positive definite; the curvature-based random-walk proposal degenerates there, producing a chain that accepts almost every draw without moving. We therefore replace the proposal with a random walk scaled to the prior, and we report results pooled over two chains of two million draws each, one started from a dispersed point; the two chains agree to the third decimal on every parameter. Under the earlier agnostic storage priors this arm was more

fragile still, with one dispersed chain settling in a region 15.8 log-points below the mode and discarded after a posterior profile showed a single well-defined optimum; under the measured priors of Section 4.4 that pathology does not recur.

The pipeline elasticity under the IEA series remains the least precisely estimated object in the paper. Under agnostic storage priors its posterior carries a long right tail reaching beyond 8; the measured priors discipline it to a 90% interval of $[0.26, 1.17]$, and we quote it with that history attached. The LNG elasticity, by contrast, is stable at 0.035 across chains and prior treatments, which is the strongest available indication that it is pinned by the data rather than by the sampler.

Results

Table 8 reports the two posteriors side by side. The change of source reverses the ordering of the two supply channels. The LNG-supply elasticity falls from 0.160 to 0.035, and the pipeline elasticity rises from 0.116 to 0.633; the posterior probability that LNG supply is more price-elastic than pipeline supply falls from 0.64 under JODI to below 0.01 under IEA, and the twelve-month variance share of precautionary demand falls from 47% to 26%, with the difference reallocated to pipeline supply. The demand-side elasticity α_{qp} is smaller in absolute value under IEA, at -0.032 against -0.098 , and the auxiliary parameters are broadly similar. The two aggregation weights move as the change of aggregate definition requires: w_L rises from 0.228 to 0.356 and w_S from 1.797 to 2.298, reflecting the different scope of the IEA supply series.

Parameter	JODI baseline		IEA robustness	
	median	90% interval	median	90% interval
γ_{qp}	+0.116	[+0.052, +0.258]	+0.633	[+0.264, +1.170]
α_{yp}	-0.015	[-0.040, -0.002]	-0.010	[-0.030, -0.001]
α_{qy}	+0.695	[+0.407, +1.013]	+0.564	[+0.281, +0.831]
α_{qp}	-0.098	[-0.256, -0.014]	-0.032	[-0.097, -0.003]
ψ_1	-0.198	[-1.202, +0.071]	-0.613	[-0.970, -0.258]
ψ_2	-0.413	[-0.881, -0.231]	-0.264	[-0.603, -0.112]
ψ_3	+0.006	[-0.107, +0.087]	-0.033	[-0.149, +0.057]
β_{qp}	+0.160	[+0.032, +0.342]	+0.035	[+0.003, +0.110]
w_L	+0.228	[+0.192, +0.265]	+0.356	[+0.273, +0.446]
w_S	+1.797	[+1.614, +1.989]	+2.298	[+2.012, +2.657]
$\Pr(\beta_{qp} > \gamma_{qp})$	0.636		0.001	

Table 8: Posterior medians and 90% credible intervals under the two data sources. Both runs use identical identifying restrictions, priors and sample. The IEA column pools three chains of two million draws; see the estimation discussion above.

We read this as a measurement result rather than as a fragility of the model, and the mechanism is the one set out above. The IEA LNG series records scheduled send-out from

terminals, a quantity determined by regasification nominations and downstream demand rather than by the spot price within the month; an elasticity near zero is what that series should deliver. The JODI series records cargo arrivals, which can be redirected toward Europe as the TTF–JKM spread widens; a larger elasticity is what that series should deliver. Symmetrically, the IEA pipeline aggregate sums gross producer-country routes with transit double-counted, a construction whose greater volatility co-moves more strongly with the price. The elasticity attributed to a supply channel is therefore a property of the measurement concept as much as of the market.

The consequence is a limit on what can be claimed about the ordering of the two elasticities. The proposition that LNG is the more price-responsive of the two channels is identified on a measurement of arrivals and does not hold on a measurement of send-out; on neither measurement is it established with confidence. For the questions posed here, whether LNG could have absorbed the pipeline shortfall, and what the binding constraint was, the relevant quantity is the volume Europe was able to attract and land, which is what the JODI series measures and why it serves as the baseline.

Figure 18 compares the cumulative impulse responses. The qualitative shapes agree across the two sources: the price response to pipeline-supply shocks builds over several months and does not unwind, the response to precautionary shocks is large on impact and decays, and the demand block behaves as in the baseline. Where the medians differ, the differences sit inside the credible bands, with the systematic exception of the responses involving the two supply channels, whose relative magnitudes follow the elasticity reversal documented above.

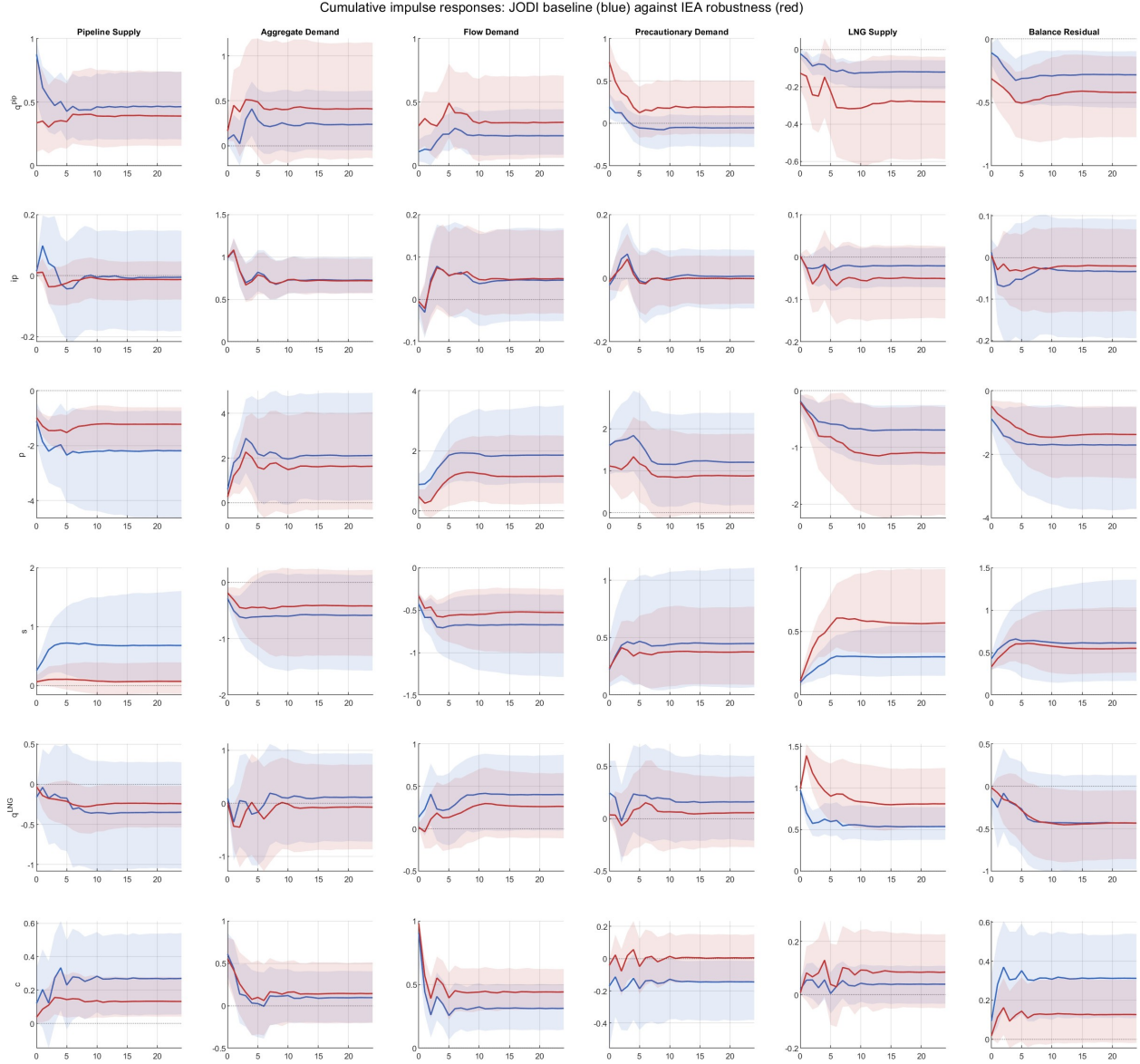


Figure 18: Cumulative impulse responses of the six endogenous variables (rows) to the six structural shocks (columns), JODI baseline (blue) against IEA robustness (red). Solid lines are posterior medians and shaded areas the 5–95% credible bands. The horizon is 24 months.